

HEALTH JOURNALISM

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BLOCK 1
UNDERSTANDING HEALTH

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BLOCK 1: UNDERSTANDING HEALTH

Introduction

Unit 1 Health: Concepts, Determinants and Dimensions: we have attempted to explain the concept, determinants, and dimensions of health. Health is a dynamic concept. Earlier health was viewed only in terms of biomedical approach. Over the passage of time many more dimensions have been added in defining the health.

Unit 2 Public Health: Genesis and Development: In terms of its genesis and development. The development of public health is also linked with the changes in the political conditions of the country. Various epidemic episodes in the country also brought reforms in the public health policy.

Unit 3 Environmental Health: Issues and Challenges: Attempts are made to explain environmental health and its related issues. The environment is the utmost concern of any society to judge the health of its people. What are the challenges encountered in the provision of good environment are elaborated.

Unit 4 Health Practices: Indigenous and Modern: This is related to the prevailing health practices in rural community. Indigenous and modern health practices are explained, along with the various components. The existing practice of medical pluralism in India is elaborated.

Unit 5 Health and Nutrition: Behaviour and Practices: Health and nutrition-seeking behaviour an important factor that defines a healthy society is explored in this unit. People in different communities, differentiates regarding consumptions of nutritional food, which in turn helps in understanding the nature and status of health seeking behaviour in rural India.

We hope this block will help in building the understanding of health among rural society which in turn depicts the development of public health system in India. How the different systems of medicine are accepted by the people which reflect the plural character in terms of indigenous and modern system.

UNIT 1 HEALTH: CONCEPTS, DETERMINATES AND DIMENSIONS

Structure

- 1.0 Introduction
- 1.1 Learning Outcomes
- 1.2 Genesis of Health
- 1.3 Concept of Health
- 1.4 Defining Health
 - 1.4.1 Critique of Definition of Health
 - 1.4.2 New Philosophy of Health
- 1.5 Determinants of Health
- 1.6 Dimensions of Health
- 1.7 Let Us Sum Up
- 1.8 Key Words
- 1.9 Further Readings
- 1.10 Check Your Progress: Possible Answers

1.0 INTRODUCTION

This unit tells about health in antiquity. Further it also tells about the concept of health. It also offers the critique of the definition of health. Further, it presents the new philosophy of health. Health does not operate in isolation, it has its determinants. This unit tells about health and its dimensions. Further it also tells about the dimensions of health. It also offers various activities which are important for the different dimension for keeping the good health.

1.1 LEARNING OUTCOMES

After studying this unit, you will be able to:

- understand the concept of health;
- enlist various determinants of health; and
- discuss the dimensions of health.

1.2 GENESIS OF HEALTH

Although concerns with health and disease have been a major pre-occupation of humans since antiquity, the use of the word 'health' to describe human 'well-being' is relatively recent. The word 'health' was derived from the old English word 'health', which meant a state of being sound, and was generally used to

infer a soundness of the body. Prior to the period of the somewhat enigmatic physician known as Hippocrates (c 460-377 BCE, or more appropriately, from around 5 BCE), health was perceived as a divine gift. Hippocrates was credited with the pioneering the move away from divine notions of health, and using observation as a basis for acquiring health knowledge. He was credited with encouraging a focus on environmental sanitation, personal hygiene and, in particular, balanced diets – “let food be thy medicine; and let thy medicine be food”. He theorized that what we currently regard as ‘health’ might be defined as the extent of a delicate balance of four fluids: blood, yellow bile, black bile, and phlegm. Ill health, he believed, resulted from an imbalance of these fluids. Nevertheless, a divine view of health persists to this era. For example, Prophet Mohammed view of health, sickness and death – followed by a high proportion of practicing Muslims – may be inferred from the following verse in the Holy Koran; “The Lord of the worlds; it is He who heals me when I am sick, and He who would cause me to die and live again” (Koran 26: 80).

With only simple technology, our ancestor could do little to improve health. Hunters and gatherers faced frequent food shortages, which sometimes forced mothers to abandon their children. Those lucky enough to survive infancy were still vulnerable to injury and illness, so half dies by the age of twenty and few lived to the age of forty (Nolan and Lenski: 1999). As societies developed, agriculture, food became more plentiful. Yet social inequality also increased, so that the elites enjoyed better health than the peasants and slaves, who lived in crowded, unsanitary shelters and often went hungry. In the growing cities of medieval Europe, human waste and other refuse piled up in the streets, spreading infectious diseases and plagues that periodically wiped out entire towns (Mumford: 1961).

1.3 CONCEPT OF HEALTH

Physicians have traditionally held a rough, commonsensical view of health as the absence of symptoms. Health, therefore becomes a residual category, the uninteresting opposite of disease. A radically different conception of health offered by one physician asserts that health is “compensated illness”. Here the usual assumption is turned inside-out. Instead of positing health as the unusual state of the human being and thinking of all illness as a deviation from some normal condition, this definition assumes that in a sense we are all ill and that our healthiness than represent an achievement in counteracting the incursion of disease. Among the attractive features of such as conception is its recognition of how pervasive illness, particularly chronic illness, is in contemporary societies and its implication that health is a process of living rather than a static entity.

The concept of health as functional efficacy enables the analyst to invoke considerations of appropriateness, of fitness for specified activities. That is, it begins to answer the question, Health for What? The health of a housewife is presumably quite different in some ways from the health of an actress; what is normally healthy in a twenty-year-old sprinter may be a state of functioning both impossible and undesirable in the middle-aged executive. If normal health consists of the ability to perform adequately in the individual’s then criteria of healthiness still vary as do the styled rights and obligations that form the

substance of roles. Health is cherished because of what it enables the person to do, and illness is feared primarily because of its interference with desired behaviours.

The notion that health reflects a dynamic patterns, changing with time and social circumstances, leads to conclusion that judgements of healthiness must be made many times as the life history unfolds. It also pushes the analyst to reiterate that most evaluations of health are relative, based on a series of perceptions and observations, rather than a unitary standard of measurement. Further, health comes in a dynamic framework to be seen as a continuum of functional ability-disability, finely graded, not as a matter of flat, mutually exclusively categories.

1.4 DEFINING HEALTH

Health is a prerequisite of human development and is an essential component for the well-being of the mankind. The health problems of the any community are influenced by interplay of various factors including social, economic and political ones. The concept of health has remained exclusive and difficult to define. Some people see health as normal condition of the individual while other view it as opposite to illness, while some consider health as well developed and adequately nourished muscular body capable of doing work and able to withstand physical strain. Thus, health can also be considered as the achievement of social and individual usefulness. The definition of health found in medical writings throughout the ages is broadly utopian. Giddens on health care in the 2nd Century, described health as a “condition of which we neither suffer pain nor are hindered in the function of daily life”, that is when we are “able to take part in government, bath and drink and eat, and do other things we want.” Health is a state in which one is able to work without any difficulty. When work is hampered, then one will be called or designated as sick. Lieban (1977) “Health and diseases are measures of the effectiveness with which human groups combining biological and cultural resource adopt to their environment. Variations in health are connected with the variations in social circumstances and habit patterns”. Leslie (1976) said that there are some universal perceptions of health and disease and at the same time there are also some perceptions unique to a particular society. When discussing health care it is important to realize that what people think as health can vary quite widely.

Health can be defined as the absence of diseases (i.e. negatively), or it can be simply as physical fitness. Another way to define health is in terms of what people are able to do (i.e. functionally), thus health may be seen as ‘optimal functioning’ or the ability to do things. It is customary to distinguish between both negative and positive definitions of health and functional and experimental definitions (Calnan: 1987). The medical view of health – the absence of disease- is clearly negative. By contrast, an example of a positive definition is that offered by the World Health Organization (WHO) (1971) has defined health as “a state of complete physical, mental and social well-being and not merely the absence of disease and infirmity”. However, comprehensive, this has rarely been practicable. Well-being is defined as a harmonious relationship between an individual or group and the physical, biological and socio-cultural environment, as also feeling of satisfaction that is associated with this. More

recently, WHO (2001) has also indicated that health is a “cumulative state, to be promoted throughout life in order to ensure that the full benefits are enjoyed in later years”. Good health is vital for maintaining an acceptable quality of life.

A functional definition implies the ability to participate in normal social roles (Parsons: 1979), and this may be contrasted with an experiential definition which takes sense of self into account (Kelman: 1975). McKenzie (2008), define health as a dynamic state or condition of the human organism that is multidimensional (i.e. physical, emotional, social, intellectual, spiritual and occupational) in nature, a resource for living, and results from a person's interactions with and adaptations to his or her environment. Therefore, it can exist in varying degrees and its specific to each individual and his or her situation. For example, a person can be healthy while dying, or a person who is a quadriplegic can be healthy in the sense that his or her mental and social well-being are high and physical health is as good as it can be (Hancock and Minkler: 2005). It is widely accepted that health status is determined by the interaction of five domains: gestational endowments (i.e., genetic makeup), social circumstances (e.g., education, employment, income, poverty, housing, crime and social cohesion), environmental conditions where people live and work (i.e., toxic agents, microbial agents, and structural hazards), behavioural choices (e.g., diet, physical activity, substance use and abuse), and the available of quality of care (McGinnis: 2001). The nature and consequences of behavioural choices are affected by social circumstances affect the health care we receive. (McGinnis et al. : 2002).

1.4.1 Critique of Definition of Health

The definition of health given by World Health Organization is also not free from the critical evaluation. The critics argue that this definition is very idealistic and does provide space means not flexible. The critics are also uncomfortable with the word “complete” as they find it is an imaginative idea that one will be completely free from all worries and without any disease or infirmity. It is unlikely that anyone would be healthy throughout his life. It is also argued that a state of complete state of physical, mental and well-being is closer to the happiness than to health (Bircher: 2005). Bircher further argues that words health and happiness shows different life experiences, where the relationships are neither fixed or constant. It implies that if one is not able to distinguish between happiness and health and if happiness is not there, which means health problem. Bircher emphasis is on changing health needs, especially in relation to age, and personal responsibility.

1.4.2 New Philosophy of Health

In contemporary time, the new philosophy of health has surfaced. We are presenting it below: Health is regarded as a fundamental human right. It is regarded as an essence of productive life and minimizing the expenditure on medical care. Health is not in the isolation rather it is inter-sectoral and regarded as essential component of social development. Health is the focal point of quality of life. It is also regarded that health is not only the concern of the individual, but it is also the responsibility of State and international society. The investment in health is investment for society and its development. Health is a worldwide and global phenomenon.

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. Explain the concept of health.

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2. Define Health.

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3. Discuss the critique of definition of health.

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4. What is the new philosophy of health?

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1.5 DETERMINANTS OF HEALTH

Health is often viewed as an individual attribute, to be experienced at personal level. It means, here the health becomes the individual responsibility and the role of society's economic, social and political system is undermined in shaping the health of the population. However, the reality is different. The role of biology, beliefs, behaviour pattern, living habits and physical environment do influences the health of the individual and in turn the health of the society. The personal attributes of the individual operate at the family and community levels. Perception of health is culturally determined or shaped by media and marketing. Financial capabilities, purchasing power of healthy food does impact the health of the individual.

Gender bias in feeding or seeking healthcare is another manifestation of prejudiced priorities at the household level. Lack of open space in terms of greenery for playing, walking, and other physical activity, poses the danger for pollution and in turn determines the health at community level. Pathways for accessing healthcare in the neighbourhood, smoke-free-public spaces, green environments and safe pedestrian or cycling lanes or crowded roads are also factors that determine good health at community level (See Figure 1.1).

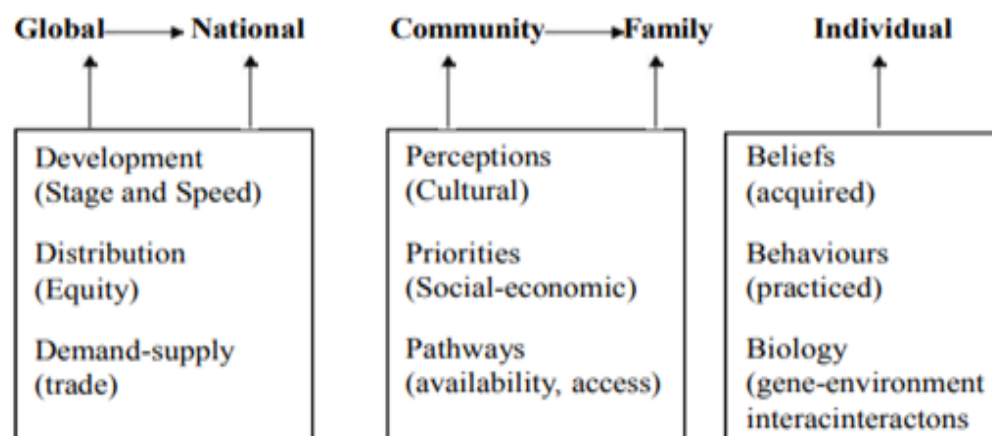


Fig. 1.1: Cascade of Health Determinants

Source: Reddy, K.Srinath (2019), *Make Health in India: Reaching a Billion Plus*, Hyderabad: Orient BlackSwan, p.2.

Further upstream are the determinants that act at the national and global levels. They involve the level of economic development and the rate at which the economy is being governed (Preston:1975). Of greater importance is how the fruits of that development are distributed across different sections of society, since inequality militates against a healthy society (Wilkinson and Pickett: 2009). Such inequality is reflected in many forms of discrimination, as society is stratified by income, gender, age, religion, caste, ethnicity, and social status, but inevitably leads to poor health outcomes, whatever the form.

At the global level, transnational trade and transport act as vectors of disease- causing agents like tobacco, and unhealthy foods and beverages, apart from pathogenic microbes (Kickbusch, Allen and Franz: 2016). Trade also influences access to medicines, health technologies and availability of nutritious foods.

There are six key elements of the health system, mainly from the perspective of a health ministry's mandate of designing, delivering and evaluating health services. These are: infrastructure of healthcare facilities; the health workforce needed; availability of essential drugs and technologies; level and use of health financing; health information systems; and overall governance of the health services. Community health focuses on people and their role as determinants of their own and other people's health in contrast to environment health which focuses on the physical environment and its impact on people's health. There are certain drivers of health in society which are delivered through health policy of the country. Social determinants are important for the health policy. At the societal level, water, sanitation, food system, physical environment, social stability and economic development determines the health status of the population. Those that operate at the individual level, although shaped by social forces, include income, education, occupation, social status, gender and participation in social networks (See Figure 1.2)

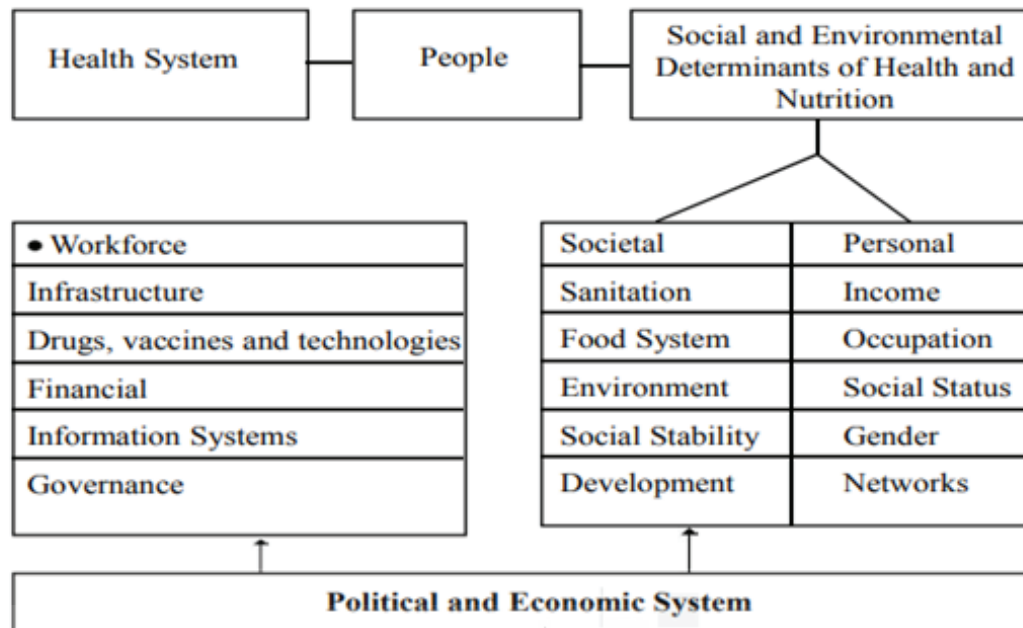


Fig. 1.2: Drivers of Health in a Society

Source: Reddy, K.Srinath (2019), *Make Health in India: Reaching a Billion Plus*, Hyderabad: Orient BlackSwan, p.4

Economic development and personal incomes exemplify the profound impact social determinism on health. The economic development of a country has a double relationship with the health of the population, with a prospering economy improving the health indicators and better health of the people, accelerating economic growth through increased productivity. Poor more likely to become sick and less likely to access or afford healthcare, while ill health pushes persons into poverty because of lost of wages and high cost of healthcare (Jamison, Summers, Aleyne et al.: 2013).

Equity becomes a very important factor when we examine the nature of economic growth or social determinants influence the health of a population. The countries with low levels of per capital Gross Domestic Product (GDP) experience a sharp rise in life expectancy when that value rises, tail an inflection point where that effect plateaus (Preston: 1975). At the same level of per capita, GDP, countries with higher income inequality have lower life expectancy and worse health indicators than countries with lower levels of income inequality within their population.

Inequality in social determinants like education, nutrition, access to safe water, sanitation and clean energy affects the health of the individuals even if health services are free available to them. An undernourished child of uneducated parents from a poor family cannot row as healthy as a child who is the well-nourished offspring of affluent and literate parents. Unfortunately, this effect can go to several generations. If pregnant is malnourished, not only will the female foetus in her womb be undernourished, but the ova of that baby, too, will experience epigenetic changes in gene expression that will manifest when that baby rows up to bear children. Both the child yet to be born and the child yet to be conceived become the victims of an intergenerational transmission of the effects of inequity (Reddy: 2016).

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) Define Food Safety. Mention the social determinants of health.

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2) How social inequality affects health?

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3) How drivers of health work in a society?

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1.6 DIMENSIONS OF HEALTH

The good health or healthy life is composed of five dimensions and these dimensions are: physical, intellectual, emotional, social and spiritual. With the passage of time many more dimensions have been added in understanding the composite nature of health. The additional dimensions are emotional, vocational, intellectual and political.

Physical Dimension

The physical dimension of the health is directly related with the perfect functioning of the body. It is related with the functioning of the body cells and organs at the optimum level. However, there is a ambiguity at the level of defining the optimum level of functioning. Some people view physical health in terms of colour of the skin, height and body weight and other physical features.

The conception of health is reflected in the form of fair colour, bright eyes, long, black and silky hair, not too fat, a good appetite, sound sleep, capacity to do hard work and no tiredness even after doing hard work. The health is also viewed in terms of well functioning of the different organs of the body and proper working of sense organs. Physical dimension of health also includes the normal pulse rate, required level of blood pressure according to age and sex of individual. Thus, we conclude that physical health refers to the state of the body, its composition, development, functions and maintenance of its vital organs. Since health is related with the functioning and maintenance of the body organs therefore, it is necessary to keep oneself healthy by doing physical exercises, eating nutritious food to keep the body and mind energized.

Mental Dimension

Mental health is not mere absence of mental illness. Good mental health is the ability to respond to the many varied experiences of life with flexibility and a sense of purpose. More recently, mental health has been defined as a state of balance between oneself and others, a coexistence between the realities of the self and that of other people and of the environment. A few short decades ago, the mind and body were considered independent entities. Recently, however, researchers have discovered that psychological factors can induce all kinds of illness, not simply mental ones. They include conditions such as essential hypertension, peptic ulcer and bronchial asthma. Some major illnesses such as depression and schizophrenia have a biological component. The underlying inference is that there is behavioural, psychological or biological dysfunction and that the disturbance in the mental equilibrium is not merely in the relationship between the individual and society. Although mental health is an essential component of health, the scientific foundations of mental health are not yet clear. Therefore, we do not have precise tools to assess the state of mental health unlike physical health. Psychologists have mentioned the following characteristics as attributes of a merely healthy person. A mentally healthy person is free from internal conflicts; he is not at “war” with himself; He is well adjusted, i.e. he is able to get along well with others. He accepts criticism and is not easily upset. He searches for identity; He has a strong sense of self-esteem; He knows his needs, problems and goals (this is known as self actualization); He has good self-control- balance rationality and emotionally; He faces problems and tries to solve them intelligently i.e. coping with stress and anxiety.

Social Dimension of Health

Man is a social animal. He is surrounded by network of social relationships. These relationships are reciprocal and satisfy various needs in the society. Various ideas and interactions take places and therefore, we also share our emotions. There are diverse cultures in the society and these cultures are regulated by various forms of norms and practices. Although, these cultures are different by they are shared and by community people and cultural diffusion also takes place. The normal individual is a part of all these function and he is capable of sharing the traits of culture and maintains the harmony. This sharing process builds the positive image and enhances the interpersonal communicational skills. This is utmost necessary to be involved in the community as well as in the society at large. The more the personal is involved in the process of integration, more he is regarded as the healthy person. Thus, the social dimension of the health includes the level of social skills, social functioning of the individual and one's ability to see one self as a member of whole society. Overall the social dimension of health is primary related with individual as a family member, he is part of society and above all he is member of larger group. It also focuses on social and economic conditions and well- being of the society which is ultimately related with the network of social relationships. There is a importance of positive human environment and positive material environment which in turn related with the social network and financial and material conditions of the individual.

Spiritual Dimension

Besides being a social being spiritualism is also necessary part of healthy life. Spiritual life makes you to turn to you goals in the life and to achieve these goals

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one strives for them. A spiritual person sets his own meanings, personal beliefs, his own acceptance or rejection of the creation. Spiritualism is related with one's inner self, therefore, there are no objective meanings attached to it. There are no universal laws which define the meaning to attain the goals, rather it is one's own understanding of existence and creation. To get released from these complexity one needs to have spiritualism which in turn affect the health. The thought of spiritualism directs the mind to think or reaches out for the constructive meaning of the life. Without considering the meaning of life one is worthless. The worth in the life ultimately transcends into good physical well-being and healthy life. This is not a old philosophy of health, rather it has emerged in the contemporary time looking to the complexity in the life. In sum, spiritualism includes integrity, principles, ethics and the purpose in life. It also includes commitment to some higher being.

Emotional Dimension

Emotional dimension of health is the domain under psychology. Since human being is not only a social being but at the same time he is also a emotional being. Therefore, emotional elements are indispensable from a healthy human being. Emotional well-being is the ability in the human being to adjust and cope with our own and others feelings. Emotions are present in all human beings in different periods of time, in different situations. At various times emotions are visible, but at times in the situation of hopelessness, depression, anxiety etc. they are not easily and therefore, can lead to the mental illness and ultimately affect the health. Therefore, one should be aware of one's weaknesses and strength which can be helpful in the coping situation of emotional disturbing situations. It will caution before hand and one can seek help so that situation can be altered. Further, it can be altered by building strong cushion of relationships with the family, peer groups and community. Emotional health is related to ones feeling whereas mental health can be seen as "knowing" or "cognition". Thus, in the present time, the mental and emotional aspects of human being should be viewed separately particularly in the context of human health.

Vocational Dimension

Vocational dimension is related with the work one does. Every human being who has come in the world is supposed to do some or other type of work for human existence. It may vary from person to person, but should be adaptable according to human capacity. Work also depends on ones capacity and limitations to perform. The performance of work is directly elated with the physical and mental health of the individual. Physical work is related with one's capacity to perform work, while its goal is associated with self realization of satisfaction and enhanced self-esteem. Its actual potential is realized only when the person is without work or he is out of work or may be retired. This situation immediately affects his health. As people think that vocational dimension is related with the economic aspects or it is values in the form of source of incomes, but it represents the persons' ability in the form of success to prove his worth in the society and does not become burden on others.

Intellectual Dimension

Intellectual dimension is related with one's ability to develop skills and knowledge to make life more meaningful. Intellectual capacity gives ability to think rationally and in turn it translates into the idea of creativity and insight in decision making. Intellectual leanings gives ability to plan the things in such a manner that will go long way and will make life successful. The mind is able to think with openness and act accordingly. It will not be influenced by any subjectivity or other external pressure to influence your decision. Additionally, positive intellectual thinking will automatically contribute to the good health. It is also helpful in the conflicting situation for arriving at the rational arguments.

A few other dimensions are also important while discussing the health. These are: philosophical, cultural, socio-economic, environmental, educational, nutritional, curative and preventive. A glance on these dimensions shows that there are many non-medical dimensions of health which are equally important for the purpose of health.

Check Your Progress: 3

Note: 1) Use the space provided below for your answers.
2) Compare your answers with those given at the end of the unit.

1) Mention the various dimensions of health?

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2) What are the social dimensions of health?

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3) Mention the importance of mental dimension in health.

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1.7 LET US SUM UP

Health is not a new phenomenon. The concept of health is a dynamic concept and therefore, its conception also gets changed. Earlier the health was viewed only in terms of biomedical approach. Over a passage of time many more dimensions have been added in defining the health. Social and preventive concept of health has expanded its boundaries to the community and environmental conditions. Health is defined as a complete fitness and away from disease. However this definition has been challenged by the critics and they view that complete health is a utopian concept and away from the reality. Health is a larger whole and

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is an aggregate of several dimensions like, physical, mental, social, cultural, emotional, spiritual, and vocational. It requires a harmonious balance of all these dimensions or in a way perfect is a balance of mind, body and spirit. Ideally, it is difficult to achieve, but efforts are needed to maximize it. In the contemporary society, the life has become so complex that one requires mental and emotional peace to counter the ill effects of complexities. Thus, to conclude we can say that health is a holistic concept and it requires a balance of all other aspects for reaming healthy.

1.8 KEY WORDS

Health: Health is a state of complete physical, mental and social well-being and not merely the absence of disease and infirmity.

Community Health: Community health focuses on people and their role as determinants of their own and other people's health in contrast to environment health which focuses on the physical environment and its impact n people's health.

Social Well-being: Well-being is defined as a harmonious relationship between an individual or group and the physical, biological and socio-cultural environment, as also feeling of satisfaction that is associated with this.

Mental Health: Mental health is a state of balance between oneself and others, coexistence between the realities of the self and that of other people and of the environment

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1.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Ref. to Section 1.3
2. Ref. to Section 1.4
3. Ref. to Section 1.4.1
4. Ref. to Section 1.4.2

Check Your Progress: 2

1. Ref. to Section 1.5
2. Ref. to Section 1.5
3. Ref. to Section 1.5 (diagram)

Check Your Progress: 3

1. Physical, Mental, Social, Emotional, Vocational, Intellectual
2. 1.6 - Social Harmony, Functioning
3. 1.6 - Response, Healthy Mental State.



UNIT 2 PUBLIC HEALTH: GENESIS AND DEVELOPMENT

Structure

- 2.0 Introduction
- 2.1 Learning Outcomes
- 2.2 The History of Public Health
- 2.3 Concept of Public Health
- 2.4 Essential Services of Public Health
- 2.5 The Development of Public Health in India
 - 2.5.1 Disease Control Phase (1880-1920)
 - 2.5.2 Health Promotion Phase (1920-1960)
 - 2.5.3 Social Engineering Phase (1960-1980)
 - 2.5.4 Health for All Phase (1981-2000)
- 2.6 Public Health and Sanitary Policy
 - 2.6.1 Early Phase of Public Health and Sanitation
 - 2.6.2 Critical Appraisal of Public Health and Sanitation
 - 2.6.3 Epidemics and Public Health
 - 2.6.4 Reforms in Public Health
- 2.7 Let Us Sum Up
- 2.8 Key Words
- 2.9 Further Readings
- 2.10 Check Your Progress: Possible Answers

2.0 INTRODUCTION

Public health promotes and protects and protects the health of the people and the communities where they live, earn, work and play. It promotes wellness by encouraging healthy behaviours. From conducting scientific research to educating about health, people in the field of public health work to assure the conditions in which people can be healthy. It ranges from vaccination to control of spread of diseases. Public health sets safety standards to protect its population and develop various health policy and programmes to ensure safety and access to healthy food.

2.1 LEARNING OUTCOMES

After studying this unit, you will be able to:

- understand the concept of public health;

- visualize the history of public health at global level and also in India; and
- discuss the public health and sanitary policy.

2.2 HISTORY OF PUBLIC HEALTH

In the beginning public health was equated with ‘the sanitary idea, and methods of preventive medicine, such as vaccination.

The development of public health started in the eighteenth century with the sanitary movement. In the early years of Second World War, public health appeared to be victorious in achieving massive reductions in mortality rates in the Western world, when scientific medicine seemed to have almost eliminated the menace of pestilence. Hippocratic texts on Airs, Waters, and Places not only represented the first systematic efforts to relate environmental factors to disease but also served as a practical guidebook. Romans produced public health knowledge and practices compatible with their social and military organization. Roman public baths made personal hygiene and cleanliness possible for all citizens.

In the Middle ages, the transmission of learning has taken by Arab scholars and physicians. In due course of time medieval cities were able to create a rational system of public hygiene for dealing with health problems,

The Renaissance is associated with the beginnings of modern science and the rise of a new urban middle class. A new science and the technologies started build up. Many of the discussions at the time resonate with contemporary ones about the balance between the role of the individual, with the feckless poor having only themselves to blame, and that of state in protecting them from danger and enabling them to make health choices.

We then move on the rise of what was called preventive medicine. This emerged in the middle of the twentieth century, coinciding with the rise of scientific medicine and optimism about its potential achievements. It was characterised by its focus on the concept of hygiene (not just in relation to infectious disease but, in some countries in the 1930s taking a rather less innocuous form with its emphasis genetic or racial hygiene, with profound implications for the ability to undertake collective health interventions in the countries concerned even now). Health professionals new best and played a key role. Education can change behaviour and many diseases can be avoided by means of mass activities such as screening.

2.3 CONCEPT OF PUBLIC HEALTH

Public health is the science of protecting and improving the health of people and their communities. Public health is defined as ‘the art and science of preventing disease, protecting life and promoting health through the organized efforts of society (Acheson: 1988). Public health promotes the welfare of the entire population, ensures security its security and protects it from the spread of infectious disease and environmental hazards, and helps to ensure access to safe and quality care to benefit the population.

The classical notion of public health, which addressed the structural determinants of health and illness, was somewhat revived by 'new public health' which gives importance to 'social' factors (Baggott: 2013). Addressing the question of ill-health of a population would thus require a political and social intervention rather than biomedical alone. Importance to social and other factors was grown out of opposition from various pressure groups, movements and critiques from medicine itself which signified that biomedical claims were no longer going to be accepted uncritically by 'lay' persons. In India new public health approach was adopted in National Tuberculosis Programme sociological data were used along with epidemiological details for diagnosis and treatment to the cases to make it as people-oriented technology. On the other hand there are criticisms about the 'New Public Health (NPH)' which claims beyond modernist strategies where it distanced itself from hospital-centred curative medicine and focus on multidisciplinary approach. NPH distances itself from biomedical approach where individual is given importance to social factors. The adoption of broad concept in its understanding includes psychological, physical and social elements. But then, NPH used medical, scientific and epidemiological knowledge for its intervention. Professional expertise like in biomedicine becomes privileged one overlay expertise in guiding the population. This kind of approach increases surveillance and enhances the power of professionals further as scope of expert surveillances, assessment and intervention is extended. The new public health has reached into the hands of technology which checks and guides the direction of public health. Technology exercises its power and knowledge by giving greater importance to technological interventions rather than societal aspects in the intervention.

2.4 ESSENTIAL SERVICES OF PUBLIC HEALTH

There are many services of public health, however, ten essential services of public health are ideal and every country should adhere to them. This framework was developed by the Core Public Health Functions Steering Committee in 1994 which included the major public health organizations at global level.

- Monitor health status to identify and solve community health problems.
- Diagnosis and investigate health problems and health hazards in the community.
- Inform, educate, and empower people about health issues.
- Mobilize community partnership and action to identify and solve health problems.
- Develop policies and plans that support individual and community health efforts.
- Enforce law and regulations that protect health and ensure safety.
- Like people to need personal health services and assure the provision of health care when otherwise unavailable.
- Assure competent public health and personal health care workforce.

Understanding Health

- Evaluate effectiveness, accessibility, and quality of personal and population-based health services.
- Research for new insights and innovative solutions to health problems.

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) Discuss the history of public health at a global level.

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2) Explain the concept of public health.

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3) Mention the essential services of public health.

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2.5 THE DEVELOPMENT OF PUBLIC HEALTH IN INDIA

India has one of the most ancient civilization in recorded history. Thousands of years before the Christian era, there existed a civilization in the Indus valley, known as Indus Valley Civilization. Mohenjodara and Harappa, showed relics of planned cities with drainage, house and public baths built of baked bricks suggesting the practices of environmental sanitation, by an ancient people as far back as 3,000 B.C. India was invaded by Aryans around 1,400 B.C. It was probably during the period, the Ayurveda and the Siddha systems of medicine came into existence. Ayurveda as a science of life developed a comprehensive concept of health. The Manu Samhita prescribed rules and regulations of personal health, dietetics and hygienic ritual at the time of birth and death, and also emphasised the unity of the physical, mental and spiritual aspects of life (Rao:1966). The post-Vedic period (600 B.C. - 600 A.D.) was dominated by the religious teaching of Buddhism and Jainism. Medical education was introduced in the ancient universities of Taxila and Nalanda, leading to the titles of Pranacharya and Pranavishara (Rao: 1966). A hospital system was developed during the reign of Rahula Sankirtyana (son of Buddha) for men, women and animals and the system was continued and expanded by king Ashoka. The origin of Unani medicine is traced back to Greek medicine. The Unani system since then became part of Indian medicine.

In the history of public health, four distinct phases may be demarcated:

2.5.1 Disease Control Phase (1880-1920)

Public health during the 19th century was largely a matter of sanitary legislation and sanitary reforms aimed at the control of man's physical environment, e.g., water supply, sewage disposal, etc. Clearly these measures were not aimed at the control of any specific disease for want of the needed technical knowledge. However, these measures vastly improved the health of the people due to disease and death control.

2.5.2 Health Promotional Phase (1920-1960)

At the beginning of the 20th century, a new concept, the concept of "health promotion" began to take shape. It was realised that public health had neglected the citizen as an individual, and that the State had a direct responsibility for the health of the individual. Consequently, in addition to disease control activities, one more goal was added to public health, that is, health promotion of individuals. It was initiated as personal health services such as mother and child health services, school health services, industrial health services, mental health and rehabilitation services. Public health departments began expanding their programmes towards health promotional activities. Since the State has assumed direct responsibility for the health of the individual, two great movements were initiated for human development, namely (a) provision of "basic health services" through the medium of primary health centres and sub-centres is an important development in the history of public health. The concept of health centre was first mooted in 1920 by Lord Dawson in England. In 1931, the League of Nations Health Organization called for establishment of health centres. The Bhore Committee (1946) in India had also recommended the establishment of health centres of providing integrated curative and preventive services. The second great movement was the Community Development Programme to promote village development through the active participation of the whole community. This programme tried to do too much too quickly with inadequate resources. It was a great opportunity lost, because it failed to survive. However, the establishment of primary health centres and sub-centres provided the much-needed infrastructure of health services especially in the rural areas.

2.5.3 Social Engineering Phase (1960-1980)

With the advances in preventive medicine and practice of public health, the pattern of disease began to change in the developed world. Many of the acute illness problems have been brought under control. However, as old problems were solved, new health problems in the form of chronic diseases began to emerge, e.g., cancer, diabetes, cardiovascular diseases, alcoholism and drug addiction, etc. especially in the affluent societies. These problems could not be tackled by the traditional approaches to public health such as isolation, immunization and disinfection nor could these be explained on the basis of germ theory of disease. A new concept, the concept of "risk factors" as determinants of these diseases came into existence. The consequences of these diseases, place a chronic burden on the society. These problems brought new challenges to

public health which needed a new phase in the 1960s, described as the “social engineering” phase. Social and behavioural aspects of diseases and health were given a new priority. Public health moved into the preventive and rehabilitative aspects of chronic diseases and behavioural problems.

2.5.4 Health for All Phase (1981-2000AD)

The glaring contrasts in the picture of health in the developed and developing countries came into a sharper focus, despite advances in medicine. Most people in the developed countries, and the elite of the developing countries, enjoy all the determinants of good health – adequate income, nutrition, education, sanitation, safe drinking water and comprehensive health care. Large numbers of the world’s people, perhaps more than half, have no access to health care at all. The neglected 80 per cent of the world’s population too have an equal claim to health care, to protection from the killer diseases of childhood, to primary health care for mothers and children, to treatment for those ill. Against this background, in 1981, the members of WHO pledged themselves to an ambitious target to provide Health for All by the year 2000, that is attainment of a level of health that will permit all people to lead a socially and economically productive life.

National Health Policy of 1983, aimed to provide Health to All by the year 2000 through the universal provision of comprehensive primary health care services. The public health services would actually need to be delivered by the state administration, NGO and other institutions of civil society. The attainment of improved health levels would be significantly dependent on population stabilization, as also on complementary efforts from other areas of the social sectors– like improved drinking water supply, basic sanitation, minimum nutrition, etc. to ensure that the exposure of the populace to health risk is minimized. This report signalled the beginning of a move away from the medicalization of public health towards one that emphasized the building of healthy public policy.

Check Your Progress: 2

- Note:** 1) Use the space provided below for your answers.
2) Compare your answers with those given at the end of the unit.

1) Mention the development of public health in India.

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2) Discuss the health promotional phase.

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3) What was the Health for All phase?

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2.6 PUBLIC HEALTH AND SANITARY POLICY

By the middle of the 18th Century, the British had established their rule in India which lasted till 1947.

2.6.1 Early Phase of Public Health and Sanitation

Before the 1860s sanitary arrangements for the civilian population were discussed sporadically. For example, in 1810, when some Dacca citizens proposed local improvements, such as the removal filth, or the repair of wells and drains, the governor-general rejected the proposal out of hand (Ahmed:1980). More concern was shown for the health of the army; for example, after 1835 medical officers were encouraged to submit details of the climate, geography, and medical statistics of districts in order to guide military campsites and cantonments to healthy places (Ramasubban:1982). However, one measure of civilian preventive medicine predated the 1860s: vaccination against smallpox was supported with fluctuating energy and attempts were made to eradicate the indigenous practice of inoculation (Arnold: 1985).

2.6.2 Critical Appraisal of Public Health and Sanitation

In 1859, A Royal Commissions was appointed to investigate the causes of the extremely unsatisfactory condition of health in the British Army stationed in India. The Commission recommended the establishment of a Commission of Public Health in each presidency and pointed out the need for the protection of water supplied, construction of drains and prevention of epidemics in the civil population for safeguarding the health of the British Army. The Royal Commission also reported that the health of the army could not be separated from the sanitary condition of the ‘native’ population” close by, and that “well-considered measures of water-supply, drainage, paving, cleansing, and general constitution in these towns would be attended with most beneficial results to the health of the troops quartered near them” All the native towns had some system of cleansing and dealing with human excreta and other nuisances; nonetheless, the commission concluded, there was much more to be done (Sanitary Commission 1865; Statistical Abstract:1870). In 1888, the Government of India directed that sanitation should be looked after by the local bodies, but no public health was created to look after sanitation. Sanitary policy was restricted to towns and military areas. For many years little was achieved in public health, partly because sanitary officers were subordinated to medical ones, and partly because sanitary concerns were restricted to official circles. In addition, technical disputes about the relationship between sanitary measures and medical science hindered the implementation of policy, especially with respect to the role of clean water in preventing cholera.

The local municipal committees focused mainly on conservancy— the removal of faeces from residential areas. But the financial limits of these committees prevented them from making such impact on sanitation. The delay in providing drainage systems has increased mortality rates. The surplus water probably formed stagnant pools, ideal breeding grounds for mosquitoes, thus increasing the spread of malaria, though deaths due to dysentery, cholera, and diarrhea probably did decline.

No rural agency existed to implement the rules established for village sanitation. Decentralization only reached the provinces in 1872, under Lord Mayo's reforms, and larger towns were given some autonomy in 1882, with the Ripon reforms. Medical administrators were uneasy about their record in sanitary work, particularly when challenged in international settings with responsibility for cholera which emanated from Bengal (Jaggi: 1979). Attempts to intervene in the provision of water, the control of sewage, and other aspects of everyday village life would probably have led to great hostility and little success. Expenditures on preventive medicine were always small, relative to overall public or to medical expenditure. Probably only water supplies and some vaccination programmes made any mark impact on morbidity and mortality before World War II.

Medical administrators appreciated the links between providing sanitation and a population's health, especially in urban areas where collective arrangements for waste disposal, water supply and drainage were major concerns.

2.6.3 Epidemics and Public Health

In 1896, severe epidemic of plague occurred in India which awakened the Government to the urgent need of improving public health. The Epidemic Diseases act was promulgated. After 1900 improvements trusts were established in many larger towns and Patrick Geddes, among others, was asked to advise on the development of urban areas— prompted partly by a concern for the appalling conditions in the slums. The trusts had power to control only new buildings, beyond the city limits. Attempts to improve working-class housing ran into overwhelming opposition from powerful slum landlords as well as the municipal committees, for whom they were too expensive.

The other impetus to urban came from plague epidemics. Plague was probably common in India before the British arrived and its association with rats was known. But it apparently disappeared from India in the early nineteenth century before being reinforced from China through Bombay in 1896. The disease spread quickly to most parts of the country, speeded by railways, trade, and commerce. The urban poor suffered most and then, panic-stricken, escaped into rural areas, taking the disease with them; Europeans were almost immune. The vital point to note here is that plague reappeared in India when control and prevention measures were understood, and yet the response of the authorities was almost totally infective. The Bombay authorities took drastic action; sufferers were compulsorily removed to hospital, the infected were segregated, premises were evacuated, a sanitary cordon was attempted around affected areas, and travellers were medically inspected. Some traders rejected bans on exports or the movement of goods, mill owners rejected steam cleansing, and jains (religious group) and others destroyed rattraps. Others concealed infected

people and then dumped their dead bodies anonymously in the street, escaped around cordons, or refused to leave infected areas for fear of theft from their unprotected houses (Klein:1973). In 1900, the plague epidemic was treated as a political emergency and regarded as a matter of public health.

2.6.4 Reforms in Public Health

During 1919 the Montague-Chelmsford Constitutional Reforms led to the transfer of public health, sanitation and vital statistics to the provinces under the control of an elected minister. This was the first step towards decentralization of health administration in India. Municipality and Local Board Acts, containing legal provisions for the advancement of public health, were passed in several provinces. The All India Institute of Hygiene and Public Health, Calcutta was established with aid from the Rockefeller Foundation. In 1937, the Central Advisory Board of health was set up with the Public Health Commissioner as Secretary and representatives from the provinces and Indian States as members, to coordinate the public health activities in the country. The Madras Public health Act was passed in 1939. In this act, the first Rural Health Training Centre was established. Remarkably in 1943, the Health Survey and Development Committee (Bhore Committee) was appointed by the Government of India to survey the existing position in regard to health conditions and health organization in the country, and to make recommendations for the future development. The committee submitted its report in 1946 and gave the recommendations of long-term programme for the attainment of reasonable health services based on the concept of modern health practice. The public health in pre-independent India thus has to be seen as a qualified failure. Even in the towns most directly amenable to British influence and action the conditions of life of most inhabitants remained appalling, with very high mortality and morbidity rates in the worst areas. Capitalist development and the rapid unplanned increase of the urban population led to slums, overcrowding, and urban squalor in India.

India became independent in 1947. Ministries of Health were established at the Centre and States. India joined the World Health Organization as a member state in 1948. Article 246 of the Constitution of India covers all the health subjects. These have been enumerated in the Seventh Schedule under three lists– Union list, Concurrent list and State list. Article 47 of the Constitution under the Directive Principles of the State Policy states; “that the State shall regard the raising of the level of nutrition and the standard of living of its people and the improvement of public health is among its primary duties”. The Planning Commission was set up in 1950 and it came out with First Five Year Plan. The Community development Programme was launched in 1952 which included all round development of rural areas.

Check Your Progress: 3

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

- 1) Mention the relationship between public health and Sanitary policy in early phase.

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- 2) Discuss the critique of sanitary provision in India?

- 3) How epidemics are related with public health?

- 4) Mention the reforms in public health.

2.7 LET US SUM UP

The development of public health started in the eighteenth century with the sanitary movement. With changes in the political conditions in India, the torch which was lighted thousands of years ago by the ancient sages grew dim, medical education and medical services became static, and the ancient universities and hospitals disappeared. In the late 1880s Florence Nightingale pressed the Indian government to improve sanitary arrangements in the villages which had been almost totally ignored. In 1888, the Government of India directed that sanitation should be looked after by the local bodies, but no public health was created to look after sanitation. Sanitary policy was restricted to towns and military areas because the remit of the Commission was to consider improvements for the towns “in proximity to military stations. The Bhore Committee (1946) in India had also recommended the establishment of health centres of providing integrated curative and preventive services. National Health Policy of 1983, in a spirit of optimistic empathy of the health needs of the people, particularly, the poor and under-privileged, had hoped to provide ‘Health for All’ by the year 2000 through the universal provision of comprehensive primary health care services.

2.8 KEY WORDS

Public Health: Public health is defined as the art and science of preventing disease, protecting life and promoting health through the organized efforts of society.

Hygiene: Conditions or practices conducive to maintaining health and preventing disease, especially through cleanliness.

Sanitation: Sanitation refers to the safe disposal of human excreta.

Epidemics: A wide spread occurrence of an infectious disease in a community at a particular time.

2.9 FURTHER READINGS

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2.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Refer to Section 2.2 18th C to Mccrecks Soc.
2. Refer to Section 2.3
3. Refer to Section 2.4

Check Your Progress: 2

1. Refer to Section 2.5
2. Refer to Section 2.5.2

Check Your Progress: 3

1. Refer to Section 2.6.1
2. Refer to Section 2.6
3. Refer to Section 2.6.3
4. Refer to Section 2.6.4



UNIT 3 ENVIRONMENTAL HEALTH: ISSUES AND CHALLENGES

Structure

- 3.0 Introduction
- 3.1 Learning Outcomes
- 3.2 Concepts and Definitions Related to Environmental Health
 - 3.2.1 Environment
 - 3.2.2 Health
 - 3.2.3 Environmental Health
 - 3.2.4 Vulnerability
- 3.3 Issues of Environment Health
 - 3.3.1 Air Pollution
 - 3.3.1.1 Ambient Air Pollution
 - 3.3.1.2 Indoor Air Pollution
 - 3.3.2 Water Pollution
 - 3.3.3 Solid Waste
 - 3.3.4 Soil Pollution and Biomagnification
- 3.4 Risk Management and Mitigation Measures
 - 3.4.1 Driving Forces
 - 3.4.2 Pressure
 - 3.4.3 State
 - 3.4.4 Exposure
 - 3.4.5 Effect
 - 3.4.6 Action
- 3.5 Let Us Sum Up
- 3.6 Key Words
- 3.7 Further Readings
- 3.8 Check Your Progress: Possible Answers

3.0 INTRODUCTION

Environmental health is one of the pressing issues in 21st century. Among the health risks, environment is becoming an important confounder that determine morbidity, mortality, and health burden of people in general and vulnerable section in particular. The poor are the most affected by low status of health by environmental factors. Nearly 30-35% burden of death is attributed to degraded environment worldwide, nevertheless, this prevalence is even higher in sub-Saharan and south Asian countries as per WHO. Environmental burden of global disease is less visible in developed countries, the situation is more precarious

in developing and least developed world. Developed countries, particularly in north America and western Europe have less prevalence of environmental burden of disease as per cent of total disease burden, which is less than 12 per cent. In this context, countries of Africa, middle east and south Asia are the worst sufferers. This is particularly because these regions are mainly comprised of poor population (Figure 3.1).



Fig. 3.1: Environmental burden of disease globally.

Source: World Health Organization, 2005.

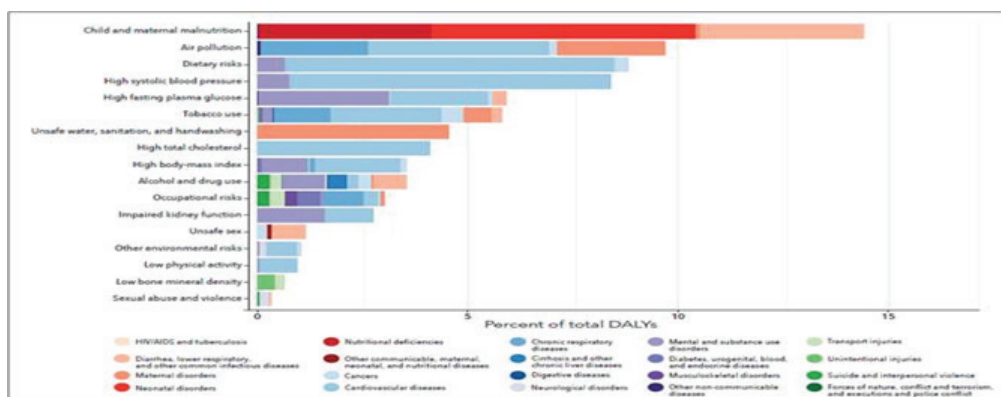
Environmental health risks range from poor air and water quality to solid waste, soil pollution, and other degraded environments.

In India, among major health risk factor between 1992-2016, air pollution was responsible for nearly 10% of all disease burden. Nearly 40% of all the diseases were non-communicable in nature. Chronic respiratory diseases, cardiovascular diseases, maternal and neo-natal disorders were driven by air pollution (Fig. 2). Unsafe water, sanitation and handwashing and other environmental risks comprised mainly of 5 per cent and 2 per cent of all disease burden, respectively. The adverse effect of unsafe water, sanitation and handwashing was visible in the form of diarrhea, maternal mortality and neo-natal mortality.

Solid waste is another serious environmental issue plaguing the urban and adjoining rural ecosystem. This poses threat to people who are in the profession of garbage collection, particularly children and women. Besides, run off from garbage during the rains pollutes the soils and water bodies in the rural as well urban areas and adversely affect the health of people.

Soil pollution and bio-magnification is another environmental issue in rural areas. Excessive use of fertilizers and chemicals is rendering the soil polluted and harmful carcinogenic substances get accumulated in the soil, and then to crops and food grown in to it. Later these substances slowly accumulate in the tissues and causes serious health issues.

Besides, there are many serious environmental issues and challenges that pose a grave risk to human health and cause mortality, morbidity and illness.



America (Figure 3.2). Source:

The combination of population pressure and economic marginalization can lead to depletion of resource base, deteriorate the existing resources and increase the level of pollution and consequent cycle of health burden and poverty. Human development index (HDI) is calculated on the basis of scores of indicators i.e. income, health and education. Given the low-income levels, the rest two indicators automatically get eroded and this becomes a cyclic in nature.

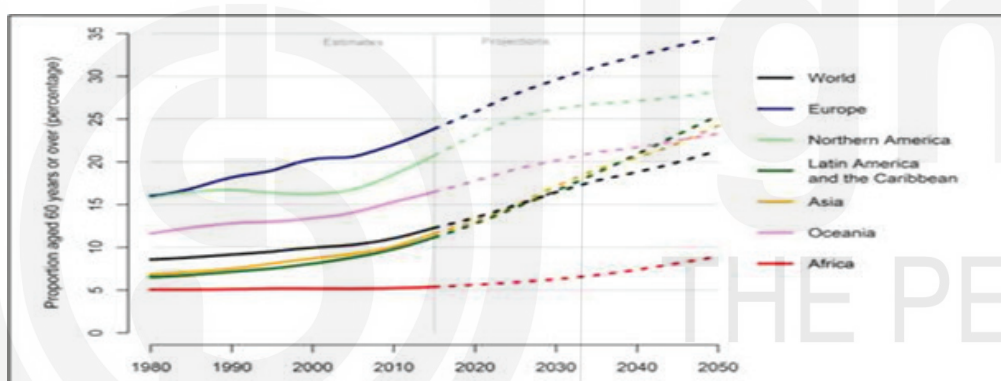


Fig.3.3: Population projection for people above 60 years Source: United Nations, 2017

Improved environment, creating sufficient knowledge base and co-ordination among the stakeholders might bring significant transformation in the health status of people across horizontal and vertical segments.

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) Write a short note on environmental health issues?

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3.1 LEARNING OUTCOMES

After reading this unit, you will be able to understand:

- difference between the concepts of environment, health, vulnerability and emerging issues and challenges;

- types of environmental health issues;
- the causes that are responsible for the occurrence of environmental health challenges; and
- possible solutions to reduce the environmental health risks.

3.2 CONCEPTS AND DEFINITIONS RELATED TO ENVIRONMENTAL HEALTH

The spike in cases of health issues due to environmental reasons like air pollution, water pollution, soil and food contamination, and sound pollution, etc. has brought our attention to understand the basic concepts of environmental health. The concepts of environment, health, well-being, vulnerability and environmental health need to be discussed and should have clear understanding. Environmental health is an integrated term implying the status of health being shaped and influenced by environmental factors. Health and well-being are two different terminologies, where the former implies the status of body parameters while the latter signifies the optimum mental and psychological status for leading a healthy life. Besides, there are many interrelated concepts like vulnerability that includes poverty, malnutrition and age sensitivity that play crucial in determining the environmental health of people.

3.2.1 Environment

While discussing about the environmental health, it is important to understand that 'environment' here refers to physical environment is a whole set of natural and social systems in which man and other living organism live and draw their necessary set of resources to survive. Environment as per World resource institute definition, environment refers to physical, chemical, and biological setting in which people live or the conditions of air, water, soil and climate. Rest others concepts like social or political environment is different from this definition. Nevertheless, it is important to understand that social environment plays an important role in determining the well-being of people.

3.2.2 Health

According to WHO, health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity. It is a multi- dimensional definition of health where physical state and mental or psychological state are combined. This unit only covers the physical aspect of health by explaining the bio-medical dimension. This refers to absence of mortality, morbidity and any disability incurred due to risk factors.

3.2.3 Environmental Health

According to WHO, environmental health addresses all the physical, chemical, and biological factors external to a person, and all related factors impacting behaviours. It encompasses the assessment and control of those environmental factors that can potentially affect health. It is targeted towards preventing diseases and creating health supporting environments. It includes the aspects of health that is determined by physical, biological, and chemical factors and the

possible solution lies in the public health efforts to reduce the risk potential and exposure. Health and survival of people is determined by the environmental factors and risks associated with it. The holistic approach is required while studying the impact of environmental factors and risks on the health. The purpose of studying environmental health is to such conditions that promote the health and prevent diseases. Understanding the environmental factors such as air, water, soil and sound is important because monitoring and assessing their levels contribute to risk aversion.

3.2.4 Vulnerability

Vulnerability is the degree of susceptibility of getting affected by the disease and reduce the mortality and morbidity. Vulnerability is determined by individual's biological as well as socio-economic traits. The biological vulnerability implies that certain age groups are susceptible to acquiring the diseases. Old people and children are at greater risk of

getting affected by air pollution related diseases such as COPD, Ischemic heart diseases, lower respiratory illnesses, bronchitis and asthma. Health of children and working mother is greatly susceptible to water borne diseases. Exposure to harmful chemical for the workers near the point source can have severe consequences. Time period to such exposure add to the vulnerability of people. While understanding the socio-economic parameters, it is important to note that poverty determine the food intake and affordability of healthcare and medicine. Poor people also live in the ecologically fragile area. They also lack resources to equip themselves with protective measures against adverse environmental conditions (Figure 3.4).

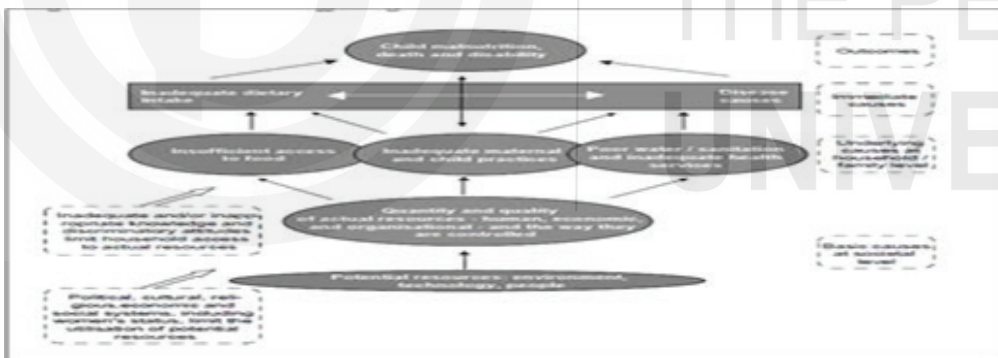


Fig. 3.4: Interplay of vulnerability factors to health

Source: Swedish International Development Cooperation Agency (SIDA), 2001.

Check Your Progress: 2

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. Define the concept of environment?

.....

2. Briefly describe the role of vulnerability in determining health?

3.3 ISSUES OF ENVIRONMENT HEALTH

Adverse environmental conditions have negative impacts on health. There are several environmental issues like air pollution, water pollution and contaminated soil, etc. with severe outcomes in case of exposure. The leading environmental issue responsible for the total health burden in 1990 was unsafe water, handwashing and poor sanitation with 12.8 per cent of total burden, followed by air pollution (11.1%) and occupational risks (2%). The situation drastically changed between 1990-2016 with air pollution becoming the leading environmental issue (9.8%), followed by unsafe water (4.6%) and occupational risks (3%). Malnutrition remains the leading risk factor to total disease burden throughout this period. This is important to note that malnutrition increases the vulnerability of individual to get infected by environmental factor in case of prolonged exposure (Figure 3.5).

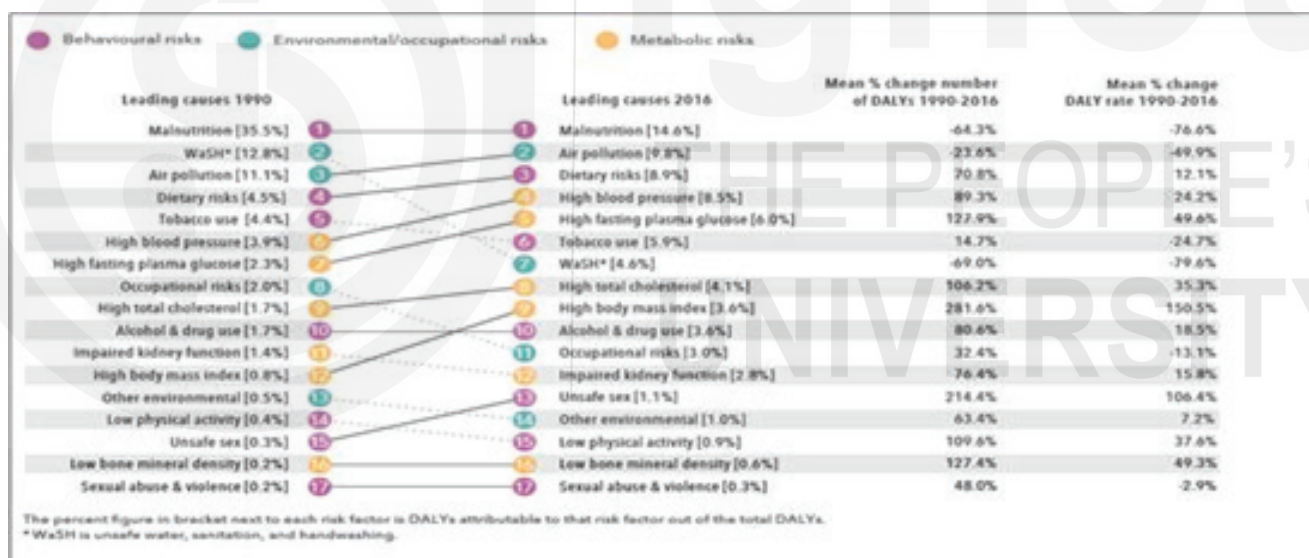


Fig. 3.5: Environmental risk factors in India among others (1990-2016).

Source: Indian Council of Medical Research, 2017.

Environmental issues of concern are broadly categorized depending upon the factor associated with it. There are many sub-categories of environment like home, outside and occupational. In the home environment, the level of exposure to concerned factor is very high. Vulnerable sections like old people and health of women and children are at particular risk. Services like water, sanitation, waste removal and ventilation need to be properly managed. Hygiene education at individual and community level through behavioural change can bring meaningful transformation.

particular risk of acquiring chemical and biological diseases, particularly to those who are at the prolonged risk of exposure. These are also the environment of high pollution load. Use of proper safety equipment and protective gears can reduce the exposure and lessen the disease burden.

Besides, the most important sub-type of environment is the outside environment. Some impact are direct whereas some are indirect as the interface between economy and ecosystem have different levels of interaction and results.

3.3.1 Air Pollution

Air pollution is the excess of harmful substances in the volume of air, becoming dangerous to human health. Air pollution is expressed in micron/meters³($\mu\text{g}/\text{m}^3$) units. There are mainly two components of air pollutants as per its nature and characteristics, i.e. gaseous and particulate matter. Gaseous air pollutants are mainly in the form of gas and mixed within the air. Many air pollutants like ammonia (NH_3), Carbon mono oxide (CO), sulfur dioxide (SO_2), nitrous oxide (NO_2), surface Ozone (O_3), smoke and chlorofluorocarbons (CFCs) are categorized as gaseous air pollutants. Particulate matter air pollutants are very fine particles like soot, dust and pollen which are rarely visible. Many of these are suspended in the air and visible only in light ray. The particles diameter varies from 1 micron to 10 microns. Depending upon its, PM_{10} and $\text{PM}_{2.5}$ are two major air pollutants. Lesser the size of particle, deadlier it the pollutant. PM_{10} can enter even in our blood veins.

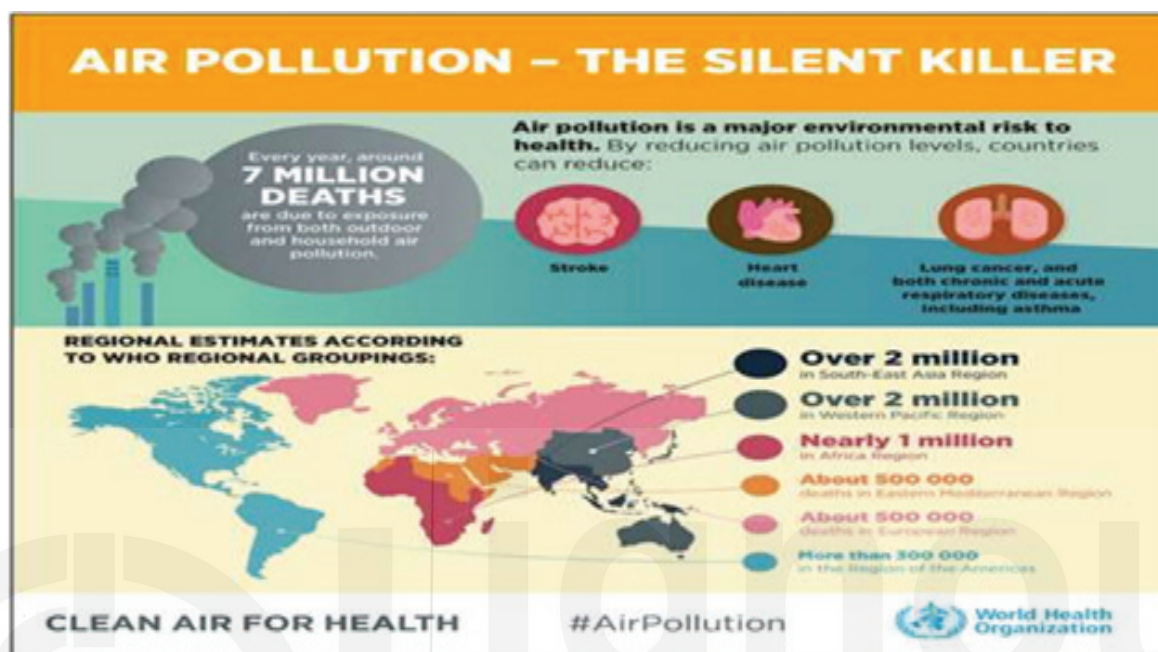
Air pollution is one of the biggest environmental threat to human health. An estimated 7 million people die each year by inhaling the polluted air worldwide as per WHO estimates. The studies show that 9 out of 10 people breathe polluted air exceeding the WHO permissible limit. Low- and middle-income countries which are in the growth transition phase, have the highest exposure. Air pollution is responsible for 7 million pre-matured deaths every year, besides, many non- communicable diseases like COPD, IHD, lung cancer and lower respiratory illnesses are the results of long-term exposure to deadly level of air pollution. The exposure to air pollution is a slow process that may take many years and kills a person slowly. As per estimates, over 2 million people die each year in the south, south-east Asian countries, followed by 2 million in pacific Asia, 1 million in Africa, 5 lakh each in Europe and Mediterranean region and 3 lakhs in Americas.

Air pollution is more severe in the urban environment of mega cities in the developing countries. The problem of air pollution and related respiratory health risks are associated with deteriorating health of urban environment and unbalanced urban land use. This situation particularly is acute in million cities where population is more than 1 million and attracting people from hinterland for various economic and social needs. While the proportion of people living in small cities is expected to decline, the million-plus cities which account for about 40 per cent of the total urban population in 2011 is expected to increase up to 47 per cent by 2025.

Thus, it is becoming a perennial challenge to control the air pollution by keeping a balance between developmental imperatives and human development Studying

the air pollution is important, particularly by integrating climatic factors and effects on human health. Based on sources and environment getting influenced, there are two types of air pollution, i.e. ambient air pollution and indoor air pollution.

Fig: 3.6: Source: WHO,



3.3.1.1 Ambient Air Pollution

Ambient air pollution is the excess of pollutants in the outer environment that is harmful for health. As per WHO, nearly 4.2 million people die each year by stroke, heart disease, lung cancer and COPD. Around 91 per cent people breathe unhealthy air worldwide that exceeds the permissible limit by WHO guidelines.

Ambient air pollution is more visible in the urban environment of developing countries like India. It comprises of both gaseous and particulate form of pollutants. Countries particularly in south Asia and Africa are struggling with the issue of air pollution. Cities, particularly mega cities in India and China have very high level of air pollution, responsible for mortality, morbidity and health burden. There are multiple sources of ambient air pollution. Fast paced developmental imperatives and low standard of technology are the major drivers for high amount of pollution. Among the sources, dust and construction account for nearly half of all pollutants, particularly particulate matters. The second biggest source of air pollution is stubble burning in the fields post-harvest and other waste burning accounting for nearly 17 per cent of all pollution. This is followed by transport (14%), emission from diesel generators (9%), emissions from industries (8%), and domestic cooking (7%). Their ratio may vary depending upon the localities within the cities (Figure 3.7).

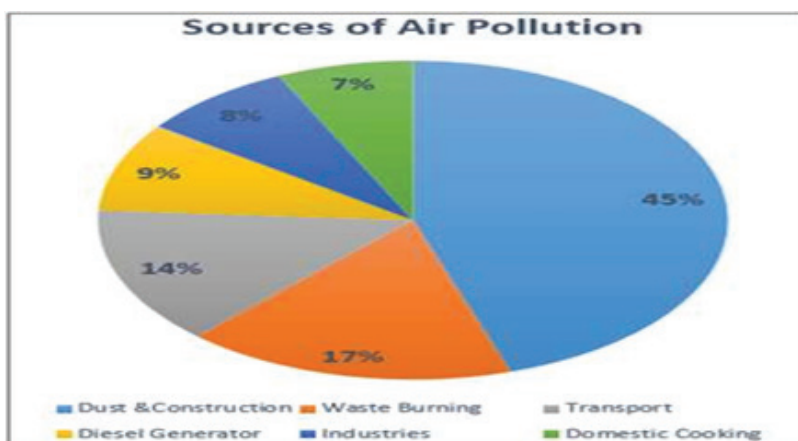


Fig. 3.7: Sources of ambient air pollution

Source: Indian Institute of Technology, Kanpur, 2015

India is home to 15 cities out of top 20 most polluted cities in the world. Delhi is the most polluted capital in the world (Figure 3.8). The situation is more severe during winters when climatic factors play their role in the high concentration. Climatic factors like windspeed, temperature and humidity during winters are less, that helps the pollutants to subside or remain stagnant in a place. Post-harvest season around November is marked by stubble burning in Punjab, Haryana and Western U.P. This generate a lot of pollutants that exceeds the permissible limits by 10-16 times and makes the air deadlier.

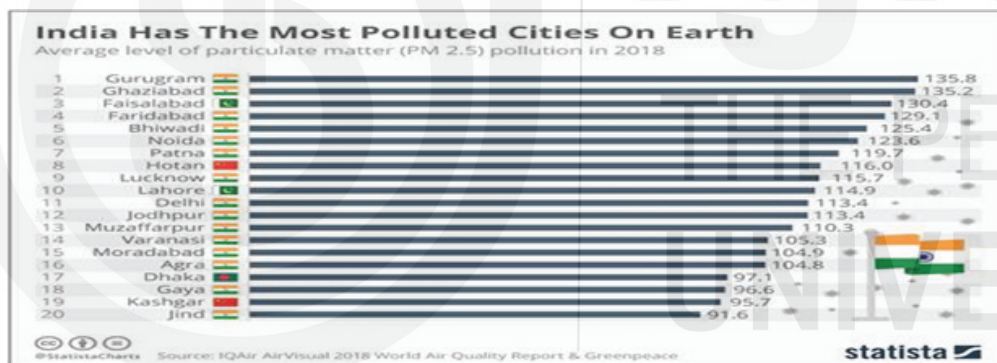


Fig. 3.8: World's top 20 most polluted cities.

Source: Greenpeace, 2018.

Central Pollution Control Board has notified the permissible limits for different types of pollutants in $\mu\text{g}/\text{m}^3$. There are different limits depending upon the type of location. For the ecologically sensitive areas, the permissible limits are less than the residential and commercial locations (Table 3.1).

Table 3.1: Prescribed Annual Limits of Major Pollutants in $\mu\text{g}/\text{m}^3$

Pollutants	Commercial and Areas Residential	Ecologically Sensitive Areas
SO ₂	50	20
NO ₂	40	30
PM ₁₀	60	60
PM _{2.5}	40	40

Source: Central Pollution Control Board (2009)

3.3.1.2 Indoor Air Pollution

Household pollution is the excess of harmful substances in the volume of air within the household. It is one of the leading causes of pre-matured deaths in the developing countries. Poor countries of Africa and South Asia like India have very high level of indoor air pollution as majority of household can't afford clean fuel for cooking and heating purposes. Fuel wood, cow dung cake and agriculture residues are the only available fuels in the rural areas of India. Though there has been a significant improvement in the access of cooking fuels including LPG, there are still a large number of people who either can't afford them or they don't have the access to them due to geographical reasons. The smoke released from cooking contains many harmful health damaging substances including particulate matter, methane (CH_4), carbon mono oxide (CO), black carbon and brown carbon, volatile organic compounds (VOC), etc. Exposure to smoke from cooking is responsible for 3.8 million deaths each year, particularly in low- and middle-income countries. Apart from the adverse health impacts, members are also subject to higher risk of burns, poisonings, musculoskeletal injuries and accidents.



Fig. 3.9 Source: google images, 2020

As per the State of Global air report 2019, an estimated 846 million people in India are exposed to indoor air pollution, that is nearly the 60 per cent of India's population. The total deaths attributed to indoor air pollution is the highest in India with 4.8 lakhs deaths per year, followed by China. Though there has been a decrease in the deaths but it is slower than China.

As per Lancet report 2015, indoor air pollution causes 1. 24 lakh pre-matured deaths per year in India. The ultra-fine $\text{PM}_{2.5}$ is mainly responsible for adverse health impact on mother and baby in the womb. Access to safe cooking fuel and alternative energy resources are critical in determining the health of family members. Women and children have the higher probability of exposure to indoor pollution than male adult members as they are mostly outside for work. In many inaccessible areas, fuel wood and cow dung are the only mode of cooking and heating. This is more acute in the hilly areas where sustainable energy is still a distant dream. Requirement of energy is also higher in the Himalayan region due to cold climate. Biogas from cow dung, fuel efficient cooking stoves and LPG are the best possible alternatives for reducing the indoor air quality.

Check Your Progress: 3

Environmental Health: Issues and Challenges

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. What are major types of air pollution? Mentions various sources of ambient air pollution?

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.....

3.3.2 Water Pollution

Water pollution is another environmental issue that causes severe health issues in public. This is more prevalent in the rural areas where sources of drinking water are polluted or potable water is unfiltered. It is the condition when the harmful substances in water exceeds the absorption capacity of water of self-cleaning. The unsafe water kills more people than any other reasons throughout the world. Sometimes it is possible to see the pollution in water by the naked eyes but mostly it remains unnoticed as it contains many harmful invisible substances, chemicals or micro-organisms degrading the quality of water and making it dangerous for human and environment.

In 2017, 29 per cent of global population unsafe managed water, including 785 million people lacking basic drinking water services as per WHO estimates. Around 2 billion people use contaminated drinking water with faeces. The drinking of contaminated water can transmit diseases like diarrhea, cholera, dysentery, typhoid and polio including other. Diarrhea is one of the very common diseases prevalent in the least developed and developing countries of Asia and Africa, that kills around 485,000 people each year globally. As the population of the world grows, the limited water resources will come under the stress each year and water stress areas will increase. As per projections, by 2025, 50 per cent of world's population will be living in water stress areas. Only the richer countries will be able to maintain the supply of clean water, in comparison to poor countries where currently 22 per cent health care facilities have no water services, 21 per cent have no sanitation services, and 22 per cent and no active water waste management services.

As per World Economic Forum, 70 % of surface water in India is unfit for drinking. Each year, 40 million liters of waste water enters in Indian rivers and other water bodies.

This is causing an estimated 9 per cent reduction in agricultural revenues each year and \$ 80 billion loss to environment and 8.7 billion cost of health. Water related diseases kills 4 lakh people each year in India. Climate change, water scarcity, population growth and industrialization and urbanization pose serious threat to existing water resources. Re-use of waste water and scientific use of water resources can reduce the pressure and pave the way for sustainable use of water resources.

There are two categories of water pollution, i.e. ground water and surface water. Ground water is a very important source of drinking water and for irrigation plain areas. It becomes unsafe due to leaching of pesticides, wastes and fertilizers through the **underground channels** that can run in to miles. Surface water is visible in the form of rivers, ponds and lakes and they are very important source particularly in poor countries and locations. Pollution of surface water also impact the agricultural production. Municipal and agricultural waste are the main contributors to surface water pollution. There are two different sources of water pollution, point source and non-point source. When contamination originated from a single source, it is called point source which include waste water discharge from oil refinery, industries treatment plants, leaking septic system, and oil spills. Non-point source of pollution is the contamination derived from diffused sources that may include agriculture and storm water runoff, debris blown into waterways, etc.

It is important to note that identification of sources of pollution is critical in cutting down on water pollution and reduce the mortality, morbidity and economic cost of health of people.



Fig. 3.10 Source: World Economic Forum, 2019.

3.3.3 Solid Waste

The threat to health from solid waste is directly related to contents and exposure level. Urban areas in developing countries, particularly in the south Asia and Africa have serious issues with solid waste management, including bio-medical wastes. Industrial waste often contains the hazardous chemicals of carcinogenic nature. Places with less sophisticated waste disposal technologies and facilities have the chance to waste materials and bio-medical get mixed up and become more lethal combination. Faecal materials also get mixed up with and become very unhygienic. Waste pickers and children playing along with at high risk of exposure. Household waste has the largest share in total waste, which is also called municipal waste. Civic bodies in developing countries like India don't have proper scientific method of collection and disposal of waste materials. Door to door collection and segregation are still not practiced in these countries except some localities within the big metropolitans. As city grows, the waste pile ups become bigger and rage of influence also gets bigger. Safe disposal of scientific disposal is costly. Less mixing of different types of solid waste

may help by adopting the practice of waste collection within the household by each responsible in different categories i.e. dry waste or wet waste; biodegradable waste or non-biodegradable wastes, etc. Besides, recycle, reuse and optimum use of materials can be very effective in generating less amount of waste materials. In the developed countries, the rate of recycling is high and there are many modern techniques of solid waste collection and safe disposal.

Problem of solid waste emanate when large portion of waste is never collected and gets piled up for many days. There begins the process of disintegration and decomposition. The foul odor fills the entire area and many harmful germs propagate within, even many of these bacteria are air borne, which makes the air unhealthy to breathe. The situation gets worse when the decomposed products seep through the soil and infects the water supply channels supplying the drinking water to people. The debris of solid waste also affect the nearby location and micro-environment like roadside, gutters and waterways. Disease bearing insects and deadly rodents find safe shelter in these debris as fly and move to streets and infects water, food and supply channels. The situation is more pathetic during rains as contamination is much faster and drainage gets clogged.

Given the poor status of waste collection, there are many informal ways of waste collection and even there is a market where children of tender age are compelled to work. This seriously damages the health well-being of this vulnerable group.

For example, exposure to deadly lead may lead to brain damage and memory loss. Delhi generate the most solid waste among metro cities in India. In Delhi, over 9,500 tonnes per day (TPD) of garbage is generated per day in the city as per Down to Earth report. About 8,000 TPD of waste is collected and transported to three landfill sites at Bhalswa, Okhla and Ghazipur. Actual waste generation in the city could be much higher, as a bulk of the waste is managed by the informal sector. According to an estimate, there are about 150,000 rag pickers in Delhi.



Fig. 3.11 Source: The Wire, 2019

3.3.4 Soil Pollution and Biomagnification

As per UNEP, 21 million hectares of land in the different locations suffers from soil pollution. In case of India, soil pollution is more visible in northern states of Punjab, Haryana, western Uttar Pradesh and Hooghly river basin in West Bengal. Soil pollution is the contamination of soil with higher amount of toxic

substances. There are many toxic substances in the soil, exposure to which can impair the health of people who consume the food grown in it. The major causes of soil pollution include excessive or improper use of pesticides, industrial waste sipping and poor management of municipal waste which runs through the drainage system and reaches to field while doing irrigation. The harmful substances in the soil reaches to our food system and finally to living tissues through the process of biomagnification. Many of these harmful substances are carcinogenic that include heavy metals, mineral oils and hydrocarbons. Toxic heavy metals like arsenic, antimony, mercury, lead, zinc, cadmium, and benzene, etc. are highly dangerous to human health. PAHs are linked to several kinds of cancers. Excessive use of pesticides and herbicides have many deadly elements to it that have a long-term impact on human health. Soil contaminants can be present in three phases. These can enter in to human body through different channels. The short-term effect to the exposure to these pollutants are headaches, nausea, and vomiting, coughing, fatigue and irritation. The exposure in long term can be very deadly as it can result in the permanent damage to nervous system and higher risk of acquiring cancers. It is very important to note that excavation of polluted soil and transporting it to the distant place is very important. Extraction of pollutants by using thermal remediation, bio-remediation and myco-remediation are other possible alternatives for reducing the level of pollutants in the soil. The regular testing of soil quality is very important to assess the health on periodic basis.

Apart from the above-mentioned environmental issues and their health impacts, noise pollution is another issue that is becoming a threat to well-being in the megacities. Climate change induced occurrence of vector borne diseases like malaria, chikungunya, dengue and kala jowar are the other adverse environmental health issues in the 21st century.

Check Your Progress: 4

- Note:**
- 1) Use the space provided below for your answers.
 - 2) Compare your answers with those given at the end of the unit.

1. What are the major health issues associated with water pollution?

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3.4 RISK MANAGEMENT AND MITIGATION MEASURES

DPSEEA framework of action is the most integrated framework of action, developed and adopted by World Health Organization for risk management and mitigation of environment health. Driving Force, Pressure, State, Exposure, Effect, and Action make DPSEEA for effective environmental health management. This is a multi-dimensional approach covering all the factors associated with environmental health. There is a detailed guidelines and associated actions to reduce the risk and take precautionary measures (Fig. 8).

Mitigations measures for particular type of environmental issue will vary from other type of issues. Therefore, it is important to adopt a holistic measure of risk management. There are following aspects of DPSEEA framework for risk management and actions.

3.4.1 Driving Forces

Driving forces for environmental health issues can vary from economic to political, social, and institutional in nature. It is necessary to mainstream the environmental health with economic development imperatives, political decisions and relating it to the institutional structure of decision making.

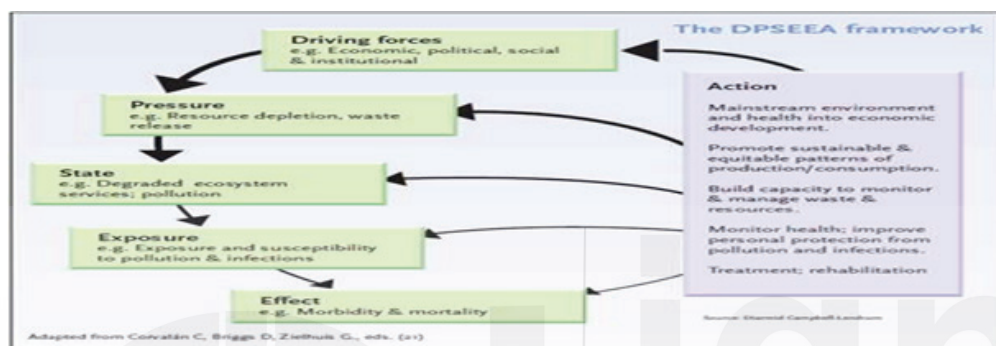


Fig. 3.12: DPSEEA framework of actions Source: adapted from World Health Organization

3.4.2 Pressure

The second important aspect of this framework is related to resource depletion and increasing pressure on natural resources. Air, water, and soil are the basic natural resources for the sustainable development. The increasing pressure induced by population explosion, urbanization and industrialization leads to deterioration of natural resource stock and pollution. Therefore, it is important to promote the sustainable and equitable patterns of production and consumption. Prioritizing the sustainable resource use and sustainable consumption pattern would significantly reduce the pressure on natural resources.

3.4.3 State

The poor state of depleted resources and pollution need to be monitored and assessed regularly to minimize the exposure. The scientific studies for evaluation of current state of pollution and degraded ecosystem services need to be tackled by building the capacity to monitor and waste and resources.

3.4.4 Exposure

Exposure is another important aspect of DPSEEA framework. Minimizing exposure is the most critical part in reducing the impact of degraded environment on human health. It also includes vulnerability of people to get exposed. Understanding the biological profile of vulnerable people, monitoring their health and improving persona protection from pollution and infections are important actions to minimize the exposure.

3.4.5 Effect

Effects are manifested in terms of mortality and morbidity. It is therefore, necessary to provide standard clinical health facilities to the people for treatment and rehabilitation. Providing lab facility for testing and lifesaving drugs is necessary to minimize the mortality rates.

3.4.6 Action

It includes a combination of actions for above mentioned pillars.

Check Your Progress: 5

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. Mention the pillars of DPSEEA framework of mitigation of environmental health.

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3.5 LET US SUM UP

After reading this unit about the environmental health: issues and challenges, you must have a good understanding about the emerging environmental issues, challenges to human health and their possible solution. You have also learnt that environmental issues are slow in nature and they are very dangerous for human health. They not only affect the health of people but also put pressure on the ecosystem services of nature. The economic cost of health burden from diseases put the family under pressure and reduces their expenditure on other things. People particularly from disadvantaged section of society like poor, old, women and children are at particular risk. Health of workers also get affected from regular exposure in their occupation. Challenges like air pollution, water pollution, soil pollution, solid waste, and vector borne diseases from climate change are of major concern today. This is more visible in the developing countries like India, where quality of natural resources is diminishing day by day. The situation gets more severe to as large number of people are poor and they lack basic health services, working in the industries and such polluted locations is also not regulated. A lot of people depend upon the informal sector for their livelihood. It is therefore, the need of the hour to integrate the environmental health planning, from top to bottom, by integrating all the factors and stakeholders for a comprehensive solution. Apart from government and policy makers, it is also our responsibility as the citizen of this country to keep our locality clean, rationalize our resource use and take care of ourselves from exposure.

3.6 KEY WORDS

Environment: It refers to physical environment with a whole set of natural

and social systems in which man and other living organism live and draw their necessary set of resources to survive.

Health: It is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.

Vulnerability: It is the degree of susceptibility of getting affected by the disease and determines the mortality and morbidity. Vulnerability is determined by individual's biological as well as socio-economic traits.

Exposure: It is the interaction of individual with environmental risk that adversely affect the health and well-being of individual.

Pollution: It is the state when the harmful substances in the natural system exceeds the dangerous limit and is enough to adversely affect the health of humans.

Mitigation: It is a pre-determined set of strategies to reduce the impact of risk factors and keep the individuals away from getting affected from environmental issues.

3.7 FURTHER READINGS

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WHO reports on environment and health.

3.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Refer to Section (pg 33-34)

Check Your Progress: 2

1. Refer to Section 3.2.1
2. Refer to Section 3.2.4

Check Your Progress: 3

1. Refer to Section 3.31 & 3.1.1

Check Your Progress: 4

1. Refer to Section 3.3.2

Check Your Progress: 5

1. Refer to Section 3.4

UNIT 4 HEALTH PRACTICES: INDIGENOUS AND MODERN

Structure

- 4.0 Introduction
- 4.1 Learning Outcomes
- 4.2 Indigenous System of Medicine (ISM)
 - 4.2.1 ISM and Medical Education
 - 4.2.2 ISM before Independence
 - 4.2.3 ISM Practitioners in Independent India
- 4.3 Medical Pluralism in India
- 4.4 Ayurveda
 - 4.4.1 Diagnosis in Ayurveda
 - 4.4.2 Treatment in Ayurveda
- 4.5 Siddha
 - 4.5.1 Diagnosis in Siddha
 - 4.5.2 Treatment in Siddha
- 4.6 Unani-Tibb
- 4.7 Yoga
- 4.8 Modern System of Medicine
- 4.9 Allopathy as a System of Medicine
 - 4.9.1 Diagnosis in Allopathy
 - 4.9.2 Treatment in Allopathy
- 4.10 Homeopathy
- 4.11 Let Us Sum Up
- 4.12 Key Words
- 4.13 Further Readings
- 4.14 Check Your Progress: Possible Answers

4.0 INTRODUCTION

From time immemorial health and disease were interpreted in a cosmological and anthropological perspective. Medicine was dominated by magical and religious beliefs which were an integral part of cultures and civilizations. Henry

Siegerist has stated that every culture had developed a system of medicine. He further states that ancient medicine was the mother of sciences and played a large role in the integration of early cultures. The organic relationship between

medicine and human advancement, any account of medicine at a given period should be viewed against the civilization and human advancement at that time. Indeed, there is a great variety of knowledge, means, methods and skills that the Indian people have gained. Medical care in India also has plurality of systems of medicine which people are using over a period of time.

4.1 LEARNING OUTCOMES

After reading the subject matter of this unit, you will be able to:

- conceptualize Indigenous and modern health practices;
- analyze various components of Indigenous health services;
- analyze various components of modern systems of medicine; and
- understand medical pluralism in India.

4.2 INDIGENOUS SYSTEMS OF MEDICINE (ISM)

Ayurveda, siddha, and unani are indigenous systems of medicine in India in relation to biomedicine that was introduced by the colonial state two centuries ago. Interestingly, homoeopathy and naturopathy– also entrants during the colonial rule– have found considerable support among people and have been formally grouped together with ISM in the current acronym of the government department (AYUSH).

Indian society is regarded as pluralistic society with different types of cultures, ethnic population and different religious categories and at the same time having rich systems of medicine which looks health, disease and causes of medicine in altogether different mode of treatment. These together are called Indian Systems of Medicine or indigenous systems of medicine. Their main idea is to look integrated health and well-being of the society. The traditional system of medicine or indigenous systems of medicine included ayurveda, unani, siddha and naturopathy. However, these systems did not receive patronage from the government and therefore, suffered from its expansion and researches. These systems of medicine are popularly known as Indian Systems of Medicine (ISM).

In 1995, the Ministry of Health and Family Welfare has given an independent identity to The Indian Systems of Medicine and Homeopathy (ISM and H). Again in 2003, these systems are renamed under the heading of Department of Ayurveda, Yoga, Naturopathy, Unani, Siddha and Homeopathy (AYUSH). AYUSH means long life. The Ministry has given directions to the state governments to include them in the recognized systems of medicine. They also opened AYUSH centres in the respective states. These systems are entrusted with better alternative for chronic diseases and also for prevention and promotive interventions for health. Besides, AYUSH, there are other health care practices which are vastly used by the rural and tribal people and they vary from one region to another region and are known as Local Health Traditions (LHT). Local Health Traditions are defined as health promotive, preventive and curative methods having mass acceptance and prevalence among household of

different socio-economic background. Although, these systems do have some threads with the indigenous health scripts, however, they do not conform to the same ancient health systems and their scripts. They may be confined to some households as home remedies or may be some services provided by folk practitioners. They are not regarded as scientific and do not have the textual references, but they are time tested through people's usability and experiential knowledge. Other forms of health providers who are informal in the social structure are also recognized by the people in the rural structure of the Indian society. They are: Traditional Health Practitioners (THP), Folk Healers (FoH), Faith Healers (FH), Dais and Traditional Birth Attendants (TBAs). They also serve the people for their health problems.

4.2.1 ISM and Medical Education

The early engagement of colonial powers with ISM was guided by economic and trade interests and a scientific curiosity supported by mercantile pluralism. Medical education in ISM has seen an enormous growth in recent years. At present, there are 219 medical colleges that offer undergraduate training in Ayurveda (BAMS), 37 in Unani (BUMS) and six in Siddha (BSMS). The total annual turnover of Ayurveda, Yoga, Unani, Siddha, Homoeopathy (AYUSH) graduates from 450 medical colleges is around 20,000. There have been some experiments in institutional arrangements for learning Ayurveda.

4.2.2 ISM before Independence

The services of vaidas and hakims were employed during the 1860s and 1870s by the administration in the Punjab to popularize its vaccination programme among the masses (Hume: 1977). Colonial state also began supporting the training of dais from 1902 onwards. The midwifery classes gradually picked up with enrolment of Christian women, and the modernizing upper and middle-class women began to use the services of the newly trained midwives while continuing to opt for home births.

The vaidas and hakims were shifting from household to bulk production of medicines at the same time that allopathic pharmacology in India was organizing itself against the total control of the British government over raw ingredients necessary for the allopathic pharmaceutical industry. The development of ISM training institutions continued with support coming from associations of ISM practitioners, wealthy patrons and princely states such as Travancore, Cochin, Gwalior, Mysore and Hyderabad. The end result was that at the time of India's Independence there were more than forty colleges of Ayurveda, Unani and Siddha (Abraham: 2005).

4.2.3 ISM Practitioners in Independent India

The establishment of various councils, national institutes, uniform curriculum, drug-testing laboratories and the publications of pharmacopoeia were direct outcomes of recommendations made by various committees set up by the government. However, a number of institutions and arrangements have proliferated in the non-state through private trusts of vaidyas, their professional associations, community institutions and NGOs that mitigate the hegemony of the scientist model.

Currently, the AYUSH sector of the Government of India is large enough to warrant a disaggregated analysis of regional trends and variations even budgetary allocation of ISM has not been significant. Firstly, the data suggest that the ISM facilities in the government institutions are uneven across the country (Abraham: 2005). For instance, they have a strong presence in states such as Maharashtra, UP, Karnataka and Kerala. These facilities are virtually absent in Jharkhand, Chhattisgarh, Arunachal Pradesh and Assam. Secondly, the data indicate that the popularity of Ayurveda, Siddha and Unani varies in different parts of the country and has also waxed and waned over the years. West Bengal, which was historically a major centre of Ayurveda, has fewer modern institutions in ISM. However, Bengal has emerged as a popular centre of homoeopathy. Thirdly, the data also contradict the general argument that attributes the popularity of ISM to the absence of allopathic facilities; those states with better distribution of allopathic facilities are also the ones with better ISM facilities. For example, in Kerala, Maharashtra and Karnataka, facilities in ISM and in allopathy are widespread (Sujatha and Abraham: 2012: 23).

According to recent data (GOI: 2005) there are nearly 7.1 lakh registered qualified AYUSH practitioners in India, of which about 2 lakhs are not institutionally qualified. The overall trend in the growth of practitioners in the ISM since 1980 shows a doubling of the figures in the last twenty years. This may be due to the sudden growth of private ayurveda colleges during this period as a result of state policies to liberalize and privatize the education sector in the country. Between 1980 and 2000, about 186 new private ISM colleges were established (GOI: 2001). Further, the number of institutionally trained practitioners has been steadily growing while the number of non-institutionally trained practitioners shows a marginal increase up to 1995 and thereafter a decline. Thus, institutionally trained ISM practitioners are taking the place held by non-institutional practitioners, in both cases far outweighing the practitioners of biomedicine.

A programme for 'Mainstreaming Ayurveda' by integrating and collocating Ayurveda, Siddha, Unani and Yoga practitioners in primary and district health centres has been launched by the Central government. It is evident that utilizing ayurveda doctors to provide allopathic services in remote areas because MBBS graduates are not willing to go there does not amount to nay recognition of ayurveda's therapeutic value.

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) What is Indigenous System of Medicine?

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2) Write down the full form of AYUSH.

.....

3) Define Local Health System.

4.3 MEDICAL PLURALISM IN INDIA

Medical Pluralism is the name given to the situation where a patient has a number of choices while selecting a system of treatment. The term was coined specifically to describe the treatment avenues available in the early modern age i.e., barbershop surgeon, apothecaries, physicians or wise women/village elders. A person suffering from illness or disease could choose from these potential treatment sources or in fact choose them all that is called medical pluralism. Medical pluralism is the adoption of more than one medical system, or the simultaneous integration of indigenous and modern system of medicine.

Pluralism in treatment-seeking among all sections of people is well known; among the poor and the well-off sections of both ‘underdeveloped’ as well as the most ‘developed’ countries (GOI: 2002; WHO: 2002).

In India, the colonial state and Independent state did not ban or strictly regulate the indigenous systems of medicine, namely, Ayurveda, Siddha and Unani (hence forth Indian Systems of Medicine ISM). In India, pluralism in health care has been officially supported since the beginning of the development of planned health services in the post-independence period, which included biomedicine and six other systems (ayurveda, yoga, naturopathy, unani, siddha and homoeopathy).

Health care in India presents a complex scenario and that is significantly influenced by colonial and postcolonial history and politics of country. Multiple medical systems such as biomedicine (allopathy), Ayurveda, Unani, Siddha, Homoeopathy, Naturopathy, Yoga and a variety of folk traditions all contribute to providing health care in present time.

4.4 AYURVEDA

Ayurveda is one of the oldest scientific medical systems in the world. The history of the origin of ayurveda starts almost from the beginning of the creation. Ayurveda is very old system of medicine in India and it has its unbroken history of over 3000 years. It also has its medical texts which are written in a period between 1500 BC to 1900 AD. Ayurveda is composed of two words – ‘Ayur’ and “Veda” which put together literally mean “science of life”. As regards the definite meaning, it would be interesting to note that the ‘life’ which is the preview of ayurveda connotes a combination of body, perceptory organs, mind and soul. The main goal of the Ayurveda is to promote health and get rid of ailments so as to enhance humanity of all sorts of pain i.e., physical, mental, intellectual and spiritual. the natural or herbal products.

For the promotion, prolongation and maintenance of health, ayurveda prescribes the observation of certain principles of daily routine, night routine, seasonal routine and ethical routine and also stresses that one must follow a regulated diet, sleep and avoidance of mental and sexual intercourse without purpose. Thus, ayurveda is not only a medical science, but also a way of life.

Ayurveda takes into account fundamental principles like the creation theory of Panchmahabhuta, the physiopathological through of Tridosha (Vata, pitta and kapha), and even the evaluation process of universe and creation, since it believes that there is no essential difference between the outside world and the human body. According to Ayurveda illness occurs if there is any derangement in the body humours such as vata, pitta and kapha or in the psychic factors such as Satwa, Rajas and Tamas caused either by excessive or inadequate interactions.

4.4.1 Diagnosis in Ayurveda

Ayurvedic diagnosis the deviation from the state of equilibrium of dosas. Ayurveda also emphasize the patient's case history, physical examination and a rational approach of diagnosis. In Ayurveda illness results from imbalance of dosas(functions). Identification of the impaired dosa/dosas play an important role in ayurvedic diagnosis. To help diagnose the pathological state of dosas, all clinical symptoms have been classified under the three dosas. To illustrate this, consider swelling. While an air-filled swelling associated with pain indicates involvement of vatta, reddishness and burning sensation denote aggravation of pitta. On the other hand, if it is a pitting/fluid-filled oedema associated with no pain or dull pain, kapha is impaired. A combination of these would indicate the derangement of more than one dosa. All clinical symptoms ranging from cold, cough, fever, and diarrhea, to swelling and skin discolouration, have been classified under vata, pitta and kapha.

4.4.2 Treatment in Ayurveda

Ayurveda defines 'cikitsa (treatment) in the following ways: 'measures aimed at removal of disease-causing factors' and 'correction of the imbalances and restoration of the health condition'. Caraka has expanded the scope of the term treatment further. He says, 'mere removal of the causative factors may not always result in complete removal of the disease because effects of the disease may still continue to be operative. So, the treatment should aim at mitigating both the cause and effect'. Ayurvedic treatment, thus, aims not only at removal of the disease-causing factors but also at restoration of the equilibrium of dosas and body functions in a way which is compatible, conducive and nourishing to the patients without weakening them. A combination of medicines, medical procedures (pancakarmas), diet and activities (mental and physical) are adopted in treatment. Just as all clinical symptoms are classified in terms of tridosas, all treatment components are also understood in terms of doshic classification. The relation between physical and mental activities and dosas are as follows: while physical exercise will increase vata, mental activities such as worrying and anger will increase vata and pitta, respectively. Everything ranging from medicines and diet to mental, physical and emotional activities and even seasons are categorized and understood in terms of the changes they cause in

the various functions and parameters in the body, that is, the dosas. Once the disease –causing dosa is diagnosed, the entire treatment is planned to bring back the deranged functions to balance using a variety of treatment modalities.

4.5 SIDDHA

Siddha system is regarded as one of the oldest systems of medicine in India. The meaning of Siddha is achievement. The literature of Siddha is written in Tamil language and is also practiced in Tamil society. This system of medicine is believed to have evolved in Tamil Nadu, although Siddha medical literature talks of the ‘Siddhas of Varanasi’. The system is largely therapeutic in nature and specializes in pharmacy. It was practiced by ‘Siddhars’, a Shaivite sect which aimed at maintaining perfect health in order to achieve ‘siddhi’ or heavenly bliss.

The practitioners of Siddha are called Siddhars. They are saintly persons and achieved the result in medicine through yoga. They establish the intense links between man and environment. The Siddhars believe that the human body is replica of the universe, as also are food and drugs. Siddha System believes that all objects in the universe are composed of five elements and these are: earth, water, fire, air and space or ether. Thus, the human body is also composed of these five elements. The food that we eat and the drugs we use are also made up of these five elements. As the earth is susceptible to natural calamities and epidemics, the human organs are influenced by food, poison, the seasons and mental stress. The Siddha system views the human body as a conglomeration of three doshas, seven dhatus and three malas. Food is the basic building material of the body which gets processed into doshas, dhatus and malas. Equilibrium between the three is considered to be ‘health’ and its imbalance regarded as ‘disease’ or ‘sickness’.

The basic principles of Ayurveda and Siddha are quite similar in terms of its fundamental and applied aspects. The basic differences which exist in these two systems is of language rather than of doctrine. Siddha medicine is more often practiced in South. Its concepts are traced back to Sangam period (500 BCE-500 CE) in India. Siddha system is very close to the chemistry and auxiliary to medicine and alchemy. The plants and minerals are regarded as of very high value for medicines in the Siddha system.

4.5.1 Diagnosis in Siddha

Siddha System believes that all objects in the universe are composed of five elements and these are: earth, water, fire, air and space or ether. Thus, the human body is also composed of these five elements. The food that we eat and the drugs we use are also made up of these five elements. As the earth is susceptible to natural calamities and epidemics, the human organs are influenced by food, poison, the seasons and mental stress. The Siddha system views the human body as a conglomeration of three doshas, seven dhatus and three malas. Food is the basic building material of the body which gets processed into doshas, dhatus and malas. Equilibrium between the three is considered to be ‘health’ and its imbalance regarded as ‘disease’ or ‘sickness’. The diagnosis of disease involves identifying its causes. This is done by examining the pulse, urine, eyes, and

studying abnormal sounds, the colour of the body and the tongue and above all, the status of agni (the digestive system of the body).

4.5.2 Treatment in Siddha

The Siddha system emphasizes that treatment is not directed towards disease but has to be in totality which includes the environmental and meteorological considerations, age, sex, race, habits, mental frame, habitat, diet, appetite, physical conditions, physiological conditions etc. The treatment focuses on normalization of the individual and it will reduce the chances of error. Siddhas basically use mercury and sulphur and also other metals, minerals, plant and mineral parts for the medicines.

Siddha experiments in yogic practices involve the use of herbal and mineral substances that mitigate the effects of severe yogic practices and aid yogic attainments. Their works on the processing of metal and mineral substances and the procedure for the correct use of herbal ingredients constitute the corpus of siddha medicine. Metals and minerals as ideals of imperishable substances became key ingredients in siddha medicine for two reasons: the idea that imperishable substances as nourishment would be more conducive to the siddha ideal of the imputrescible body than herbs that are perishable, and the need to ensure steady supply of medicine unfettered by the seasonal availability of herbs.

4.6 UNANI-TIBB

The word unani or (Yunani) is a corruption of the Arabic word for Ionian (i.e. Greek). The unani system is originated in Greece and in India also it has its very strong roots. Around in eleventh century, it came to India and introduced by Arabs and Persians. It was subsequently indigenised in the Indian subcontinent through interchange with ayurveda and through the translation of Arabic medical texts into Persian and Urdu, acquiring the name Yunani tibb (Kumar: 1997; Alavi: 2007; Attewell:2007). Unani medicine came from West Asia eight hundred years ago.

This system believes that disease is a natural process and the function of the physician is to aid the natural forces to the body. Unani system is based on a humoural theory. The four humours of this system are in the form of: blood, phlegm, yellow bile and black bile. Temperament is assigned to each humour and they are as follows: blood is hot and moist; phlegm is cold and moist; yellow bile is hot and dry; and the black bile is cold and dry. The drugs in the unani system are also assigned temperaments. These temperaments are: every person has a unique humoural constitution which represents his healthy state. Any change in this brings about a change in his state of health. A person also has a power of self-preservation or adjustment which strives to restore disturbances within his constitution. Unani places great reliance on his power. Thus, unani treatment not only helps the person to overcome the present disturbances, but also facilitates the person to acquire additional power of resistance to further disturbances. Since humours are produced for digested food, unani practitioners attach great importance to diet and digestion both in health and disease. For diagnosis unani relies heavily on its methods of testing pulse. Unani system

asserts that the body has a self-preservation power and therefore, it always tries to restore any disturbance within the parameters prescribed by the constitution or state of the individual.

4.7 YOGA

Yoga is a scientific system which focuses on the interaction of mind, body, brain and behaviour. It is basically to use the mind which in turn will affect the physical functioning and results in promoting the health. Its basic method is mediation which includes certain postures, directed attention and free mind and does not involve any distraction. People use mediation to increased their focused attention, calmness and relaxation so that they can cope with illness and able to enhance their health and well-being. There are various types of yoga practices for various types of ailments. Yoga is a combination of physical postures, breathing techniques and mediation. Yoga is also regarded as an art of living which focuses on persons physical, moral, mental and spiritual aspects. This is not the propriety of certain persons but it is practiced across the globe by different race, class, age, sex and creed for the well-being of the society.

The meaning of yoga is joining or yoking the individual soul to the universal soul which means the union of the personal soul with the God. The origin of yoga word is from the Sanskrit word 'yuj' meaning thereby 'to join' or 'to yoke'.

Yoga practices are regarded as very old healthy practices. Yoga asana are also narrated in Rigveda. Upansihads do have different mentions of yogic exercises and procedures. Patanjali was the one who systematized the existing yoga practices around third century BC. He also linked to the Samakhya School of Philosophy. Thus, Patanjali evolved the yoga philosophy and then enunciated in his Yoga Sutras. He mentioned eight components of yoga namely: yama, niyama, asana, pranayama, pratyahara, dharana, dhyana, and samadhi. Yama, niyama, asana, pranayama and pratyahara are related with the body and they prepare the body for the dharana, dhyana and **samadhi** which are related to the mind. The basic tenets of yoga are: restraining of social behaviour (yama), good personal conduct (niyama), Yogic exercise (asana), breathing exercises (pranayama), mediation (dhyana), and attainment of super-consciousness (Samadhi). Yoga does not involve any chemical, physical or botanical agent which is administer in the body in the form of medicine. It is accepted as a global medicine for keeping the health of the people. Baba Ramdev is regarded as the great practitioner of yoga in the country and is very popular with the yogic exercises and also benefits of yogic exercises with minimal effort (Danik Jagran: 5th Sept., 2005).

Check Your Progress: 2

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) What is Medical Pluralism?

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2) What is Ayurvedic system of medicine?

3) Write down the basic tenets of yoga?

4.8 MODERN SYSTEMS OF MEDICINE

Hippocrates (460 BCE-377BCE), considered the father of modern medicine, brought to its realm the concept of logical rather than supernatural explanations for illness. The central tenet of his theory was that illness was a result of imbalance among the four humours– blood, black bile, yellow bile and phlegm. Centuries later, Galen (130-201 CE), a Greek philosopher and physician, stressed the importance of anatomy but studied it using animals, mainly pigs. He extrapolated human anatomy from that of pigs, whose anatomy he considered similar to humans. In 1539 CE, following the judgment by an Italian judge, Vesalius was able to dissect the bodies of executed criminals for studying human anatomy. Structures that were previously extrapolated from those of animals and only imagined were now being visualized. (Ventura: 2000; Nutton: 2004).

In addition, William Harvey's discovery in 1628 CE of the circulation of blood by the heart marked the beginning of an era of viewing the human body as a mechanized assemblage of organs supplied with fuel/energy by the blood. The use of microscope by Robert Hooke in 1664 CE to view cells marked yet another stage of development in the field of western medicine. Information about microscopic pathological anatomy played a crucial role in making the transition of western medicine from philosophy to science.

The industrial age brought in new diseases and to meet this challenge, the physicians discarded the classical theories of Hippocrates and Galen and adopted newer concepts such as viewing symptoms as manifestations of the disease. This also led to the concept of treating diseases by suppressing their symptoms, which continues to date (Nutton: 2004).

Nowhere health delivery system or usability of health care systems is exclusively based on one system of medicine. In fact, there are varieties of systems of medicine which have evolved over a period of time in different parts of the world. These systems of medicine have a differential growth in relation to its origin and scientificness. How these different systems of medicine have survived and grew finds its answer in the social and historical conditions of the

society in which they live. There is an evidence that every society uses multiple systems of medicine. They are not exclusively dependent on one system of medicine. With the forces of globalization, the world is rapidly free flowing of different systems of medicine. People have varieties of system to choose for different ailments different systems of medicine. India has, perhaps, the world's largest community supported living tradition of its own indigenous systems of medicine.

The western systems of medicine include Allopathy and Homeopathy is included in other systems of medicine below we will discuss different system of medicine prevailing in India.

4.9 ALLOPATHY AS A SYSTEM OF MEDICINE

Allopathic medicine is an expression commonly used by homeopaths and proponents of other forms of alternative medicine to refer to mainstream medical use of pharmacologically active agents or physical interventions to treat or suppress symptoms or pathophysiologic processes of diseases or condition. The expression was coined in 1810 by the creator of homeopathy, Samuel Hahnemann (1755–1843). Never accepted as a mainstream scientific term, it was adopted by alternative medicine advocates to refer pejoratively to mainstream medicine. In such circles, the expression “allopathic medicine” is still used to refer to “the broad category of medical practice that is sometimes called Western medicine, biomedicine, evidence-based medicine, or modern medicine”.

The term allopath was used by Hahnemann and other early homeopaths to highlight the difference they perceived between homeopathy and the medicine of that time. Practitioners of alternative medicine have used the term “allopathic medicine” to refer to the practice of conventional medicine in both Europe and the United States since the 19th century. The term allopathic was used throughout the 19th century as a derogatory term for the practitioners of heroic medicine a precursor to modern medicine that did not rely on evidence.

Allopathy medicine is what most of us understand to be “standard” or “regular” medicine as practiced and distributed by the practitioners. We recognize it as legitimated, true and credible. In broad definition, “Allopathic medicine is that practice which combats by use of remedies producing effects different from those produced by the disease treated, including the use of all measures that have proved to be some value in the treatment of disease” (Rosengren: 1980). Thus, from such a view, the human body in its normal state is free of disease, and therefore, any disease found must be regarded as a foreign intrusion into an otherwise healthy organism. Further, allopathic medicine is rooted in the other general idea that in as much as disease is foreign its cure can usually be brought by the application of some form of opposites to it.

Scientific medicine is characterised by the assumption that (1) all disease is martially generated by specific etiological agents such as bacteria, viruses, parasites, genetic malformations, or internal chemical imbalances; (2) a passive patient role; and (3) the use of invasive manipulation to restore/ maintain the human organism at a statistically derived equilibrium point (health). It is based

itself around the discovery of micro-biological agents (Bacteria) as the cause of disease and around the theory of scientific etiology as a mechanism for explaining the role of these agents. Scientific medicine was not just a new name for allopathy, although many allopaths adopted the framework and terminology of the new medicine as an attempt to appropriate the legitimacy conferred the name scientific medicine. Scientific medicine has a different theoretical basis and epistemology than other existing modes of medicine of late nineteenth century, but it has an extremely limited clinical and therapeutic repertoire (Berliner: 1983).

4.9.1 Diagnosis in Allopathy

Diagnosis in allopathy generally begins with an assessment of the patient's case history followed by physical examination and a number of laboratory tests ranging from simple laboratory evaluations to sophisticated imaging modalities. The emphasis is on objectivity in diagnosis using high-end technologies and various laboratory techniques. What is considered normal within its conceptual framework is known through various laboratory parameters and the patient's condition is evaluated against those accepted as 'normal' parameters— any deviation from normal values is considered to indicate a disease state.

The diagnostic techniques generally structural abnormalities both at grass-root level and subtle levels. For example, imaging techniques such as X-ray, ultrasound, computerized tomography (CT) and magnetic resonance imaging (MRI) provide information about structures and structural abnormalities such as fracture in a bone and cancerous growth in an organ. The biomedical rests also focus on structural entities, though at a subtler level. For example, anemia is diagnosed by decreased hemoglobin and diabetes by the levels of glucose and insulin, all of which are subtler structural entities. Microbiological tests also involve identification of structures like bacteria, viruses or parasites. Although there are techniques for providing functional information such as electrocardiogram (ECG), electroencephalogram (EEG) and electromyogram (EMG), all of which measure the electrical activities of tissues, the focus is generally on structural abnormalities. Many of the diagnosis techniques are designed and fine-tuned to observe structures and structural changes which need not be the causative factors of the disease (Jayasundar: 2012).

4.9.2 Treatment in Allopathy

It is well-known that a substantial number of diseases in biomedicine are regarded as 'idiopathic'. In the absence of the knowledge of causation, the treatment is quite often symptomatic and interventional. Surgery plays a major role in treatment and is used to rectify or replace damaged structures. Drug interventions include supplementing deficiencies (for example, iron in anemia, and insulin in diabetes) and manipulating the chemistry of body. Consider the example of osteoarthritis- it is diagnosed with imaging techniques such as X-ray, CT or MRI, all of which give structural information pertaining to the affected joint/joints. The clinical symptoms are generally pain and loss of movement. Although obese patients are generally advised weight reduction and exercises, the first line of treatment is suppression of pain with painkillers. Surgical intervention is advised for patients who have severe pain despite the use of

painkillers (American College of Rheumatology: 2000). While symptoms are evaluated thoroughly and objectively, the etiology of symptoms is not known. Since the root cause is not diagnosed, all treatment strategies aim at reducing the symptoms.

4.10 HOMEOPATHY

Homeopathy as a system of medicine has slowly but definitely gained a foothold in India. It is based on the principle of ‘similia similibuscurentis’, namely whatever causes the derangement of the vital force alone can cure it best. In this lies the strength of the system. The second outstanding feature of homeopathy is that it treats each individual as a separate identity and the total personality—mental and physical—as one whole. It believes that an individual’s state of mind has a direct bearing on the health of the body. The mental symptoms or how the individual feels is taken to be the most important guiding force. Weather, the time of day and night is also taken into consideration while prescribing treatment. In homeopathy, importance is given to the part played by food and way of life in curing a disease. In India, homeopathy is popular in all major cities; in West Bengal it receives its highest patronage. Today, there is a resurgence of interest in traditional medicine. Ironically, the inspiration for this has arisen out of desperate search for alternatives to allopathic medicine and has resulted into yoga, meditation, acupuncture etc.

In 1810, some German physicians and missionaries arrived in India and they brought Homeopathy. They distributed their homeopathic remedies in Bengal.

The Government of India, established the Central Council of Homeopathy. More than a century and a half, Homeopathy is practiced in India. It is regarded as one of the important system of medicine in providing health care to the number of people in the country. Its importance lies in viewing the person in its totality and looks at the inner balance at mental, emotional, spiritual and physical levels.

Check Your Progress: 3

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) What is modern health practices?

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2) Define Allopathy.

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- 3) Write down full form of CT, MRI and ECG.

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- 4) What is the main principal of Homeopathy?

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4.11 LET US SUM UP

India was invaded by the Aryans around 1400 BC. It was probably during this period the Ayurveda and the Siddha system of medicine come into existence. Ayurveda or the science of life developed a comprehensive concept of health. The Manu Samhita prescribed rules and regulation for personal health, dietics and hygiene ritual at the time of birth and death and emphasized the unity of physical, mental and spiritual aspects of life. The next phase in Indian history (650-1850 AD) witnessed the rise and fall of Mughal empire. The Muslim rulers introduced the Unani system of medicine sometime around 1000 AD. The Arabic system of medicine, popularly known as unani system of medicine is traced back to Greek medicine. With changes in the political conditions in India, the torch which was highlighted thousands of years ago by the ancient sages became static, and the ancient universities and hospitals disappeared. By the middle of the eighteenth century, the Britishers brought the allopathic system of medicine in India, which is based on the scientific theories of science. This system has achieved a great deal of success in curing illness.

4.12 KEY WORDS

Ayurveda: Ayurveda is basically a knowledge about life and deals with the humanity. Its treatment is based on the natural or herbal products.

Unani: A system of medicine which originated in Greece and believes that disease is a natural process and the function of the physician is to aid the natural forces to the body.

Siddha: A system of medicine which is therapeutic in nature and specializes in pharmacy. It aimed at maintaining perfect health in order to achieve 'siddhi' or heavenly bliss.

Homeopathy: Homeopathy as a system of medicine is based on the principle of 'similia similibuscurentis', namely whatever causes the derangement of the vital force alone can cure it best. It gives importance to the inner balance at mental, emotional, spiritual and physical levels.

Yoga: Yoga is a scientific system which focuses on the interaction of mind, body, brain and behaviour. Yoga is also regarded as an art of living which focuses on persons physical, moral, mental and spiritual aspects.

4.13 FURTHER READINGS

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4.14 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Refer to Section 4.2
2. Refer to Section 4.2 (pg 52)

Understanding Health

3. Refer to Section pg 52-53

Check Your Progress: 2

1. Refer to Section 4.3
2. Refer to Section 4.4
3. Refer to Section 4.7

Check Your Progress: 3

1. Refer to Section 4.8
2. Refer to Section 4.9
3. Refer to Section pg 62 (4.9.1)
4. Refer to Section 4.10



UNIT 5 HEALTH AND NUTRITION: BEHAVIOUR AND PRACTICES

Structure

- 5.0 Introduction
- 5.1 Learning Outcomes
- 5.2 Health Scenario in Rural India
- 5.3 Determinants of Health Seeking Behaviour
- 5.4 Impact of Rural Health Services
- 5.5 Health Seeking Behaviour Due to Technology
- 5.6 Alternative Medicine and Rural Health
- 5.7 Barriers to Health Seeking Behaviour
- 5.8 Challenges in Rural Health
- 5.9 Solutions to Rural Health
- 5.10 Let Us Sum Up
- 5.11 Key Words
- 5.12 Further Readings
- 5.13 Check Your Progress: Possible Answers

5.0 INTRODUCTION

Health and nutrition have an immense effect on the lives of humans. Good nutrition is an important factor to lead a healthy life. A certain amount of nutrition is needed from an early age for the proper development and growth of the body. Various kinds of foods have different amounts of nutrition and the intake of nutrition depends on the several factors like the age, sex, weight etc.

Better nutrition is related to improved infant, child and maternal health, stronger immune systems, safer pregnancy and childbirth, lower risk of non-communicable diseases (such as diabetes and cardiovascular disease), and longevity. Healthy children can learn better. Also, people with adequate nutrition are more productive and can create opportunities to gradually break the cycles of poverty and hunger.

Malnutrition, in every form, presents significant threats to human health. Malnutrition refers to deficiencies, excesses, or imbalances in a person's intake of energy and/or nutrients. Today the world faces a double burden of malnutrition that includes both undernutrition and overweight, especially in low-and middle- income countries. Every country in the world is affected by one or more forms of malnutrition. Combating malnutrition in all its forms is one of the greatest global health challenges.

Health and nutrition seeking behaviour an important factor that defines a healthy society. People's behaviour differentiates regarding consumptions of nutritional food. The WHO indicates consumers preferences and intra-household decision making as one of the factors related to vegetable consumption in sub-Saharan Africa. Also, it differs from rural to urban, male to females, poor to rich, etc. There are many factors that determine people's behaviour towards health. Inappropriate health seeking behaviour has been linked to worse health outcomes, increased morbidity and mortality and poorer health statistics. Hence, it is important to maintain an appropriate health and nutrition seeking behaviour and practices among the people to improve the health status of a nation and thus its economic development.

5.1 LEARNING OUTCOMES

After reading this Unit, you will be able to:

- understand the nature and status of health seeking behaviour in rural India;
- learn the importance of alternative medicine; and
- identify the gaps between health service availability, accessibility, use and reasons for the same.

5.2 HEALTH SCENARIO IN RURAL INDIA

India is the second most populous country in the world and has changing socio-political demographic and morbidity patterns that have been drawing global attention in recent years. Despite several growth orientated policies adopted by the government, the widening economic, regional and gender disparities are posing challenges for the health sector, especially in rural India. Among 195 nations with more than a billion people, India has been ranked 145th on the Health Access and Quality Index making it one of the biggest underachievers in Asia behind its neighbours like China, Bangladesh, Sri Lanka and Bhutan, according to a Lancet study.

India is in the limelight at the global front not only in terms of population burst but also in its health scenario. Even after celebrating its 73 years of independence, its population is still brewing under the threat of a degraded health system. There are approximately 85 per cent of the populations who are still fighting for basic healthcare services in their area. This situation has been promoted by worsening living conditions of rural habitats. It is observed that 70 per cent of population has no access to specialist care as 80 per cent of specialists live in urban areas. Only 13 per cent of the rural population have access to primary health centers, 33 per cent to sub-center and 9.6 per cent to a hospital (NFHS-II). The prevalence of underweight children was higher (38%) in rural areas compared to urban cities (29%). Only about 10 per cent children under the age 6-23 months were reported to receive an adequate diet. This inequality in access is accentuated by the stark state level disparity in nutritional status.

The unhygienic and unhealthy conditions of household, unsafe drinking water, open defecation, aggravate expansion of several diseases in these areas. The scenario gets worse through the superstition practiced by rural people. The

blind faith of tribals that any disease may be cured by magic has subjugated the minds of the rural population of India. Due to this kind of impression, the rural areas are under the influence of various malpractices which ultimately seal off the procession of modern pathology here. One of the important reasons for the decline of rural health care is inadequate human resources in the health care system. The health institutions like Primary Health Centre (PHC), Sub-Centre (SC), and Community Health Centre (CHC) are facing huge problems because health professionals are absent. Doctors don't want to work in rural areas either because of infrastructure inadequacy or lack of incentives. In a study conducted by Banerjee, in Rajasthan around 45 per cent of doctors were absent from PHC and 56 per cent from SC. The condition gets intensified with not or little qualified practitioner, minimal amount of expenditure on public healthcare which counts to be 17.9 per cent of total expenditure (Bhat, Ramesh and Nishant Jain, 2004). In comparison to public funding, private funding (private household health expenditure, charity, funds from NGOs, etc.) contribute a great chunk in healthcare.

However, some progress has been made since independence in the health status of the population; this is reflected in the improvement in some health indicators. Under the cumulative impact of various measures and a host of national programs for livelihood, nutrition and shelter which included various steps taken by the government to improve the health scenario in rural areas. Several strategies and missions have been initiated for institutionalizing the prevailing rural health framework to uplift the health standard of common mass. Some of the programmes introduced by the government focusing rural health are:

National Rural Health Mission (NRHM): National Rural Health Mission (NRHM) has been one of the central achievements in the field of rural healthcare. It was first initiated in the year 2005 with an objective to deal with the problems and feebleness across primary healthcare and enhance the status and system of rural areas. It provides effective, accessible, accountable, inexpensive and reliable healthcare to the mass and in particular to those sections who are more poor, vulnerable and prone to health disease.

Pradhan Mantri Matritva Vandana Yojana (PMMVY): PMMVY, previously known as Indira Gandhi Matritva SahyogYojna (IGMSY), introduced in the year 2010, is a maternity program run by the government of India with an objective to encourage women to follow Infant and Young Child Feeding (IYCF) practices including early and exclusive breastfeeding for first six months. It is a conditional cash transfer scheme which is implemented through the State ICDS Cells for pregnant and lactating women of 19 years of age or above for first two live births, from the Ministry of Women and Child Development. It provides partial wage compensation to women for wage-loss during pregnancy, childbirth and childcare.

Janani Suraksha Yojana (JSY): Janani Suraksha Yojana is a flagship program launched in 2005 under the National Rural Health Mission (NRHM) of Government of India modifying the National Maternity Benefit Scheme (NMBS). It is intended to promote institutional delivery to reduce maternal and neonatal mortality. It provides cash incentives to the women for delivering their child in government or other private medical facilities over home-based

Health Insurance through Rashtriya Swasthya Bima Yojana (RSBY): Rashtriya Swasthya Bima Yojana (RSBY) is one of the important schemes in the area of rural health. It was launched in the year 2008, which was earlier designed to target only the Below Poverty Line (BPL) households, but has been expanded to cover other defined categories of unorganized workers. Its objective is to provide financial aid for households affected by major health shocks and improve health outcomes (Das, Jishnu and Jessica Leino, 2011).

Swachh Bharat Abhiyan (SBA): Swachh Bharat Abhiyan or Clean India Mission is a campaign in India initiated in the year 2014, whose objective is to eliminate open defecation through the construction of household-owned and community owned toilets. Still in many parts of rural India open defecation is practice which causes the most common life taking diseases like diarrhea, typhoid, hepatitis, intestinal worm infections, cholera, etc.

Even though there were many programmes most of them lacked proper implementation hence, couldn't achieve the desired goal. India being the nation of villages requires an intensive access to rural health care. Rural health is one of the important subject matter to rural life and It's the right of every individual but lack of quality infrastructure, dearth to medical facilities and shortage of primary health care components are denying that right to them. Hence, health care in rural India is a major challenge for the policy makers which should be dealt very effectively and seriously.

Check Your Progress: 1

- Note:**
- 1) Use the space provided below for your answers.
 - 2) Compare your answers with those given at the end of the unit.
- 1) List the significant programmes by the Government of India to improve the health status in Rural India.

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5.3 DETERMINANTS OF HEALTH SEEKING BEHAVIOUR

Healthcare seeking behaviour (HSB) has been defined as, “any action or inaction undertaken by individuals who perceive themselves to have a health problem or to be ill for the purpose of finding an appropriate remedy”. Health seeking behaviour can also be referred to as illness behaviour or sick-term behaviour. Health seeking behaviour is situated within the broader concept of health behaviour, which encompasses activities undertaken to maintain good health, to prevent ill health, as well as dealing with any departure from a good state of health.

Studies suggest that several factors influence the HSB of the population and certain segments of the population are more likely to use appropriate HSB than

others. Inappropriate HSB and its effects have been found to be skewed among different population segments. For example, in Pakistan, households whose average income was below the minimum wage were less likely to seek formal medical care for their illness than those whose incomes were above the minimum wage. In Kenya, almost 70 per cent of pregnant women within households in the upper socio-economic stratum were found to have their deliveries in health facilities compared with 42 per cent among pregnant women in the middle socio-economic stratum and 38 per cent in the low socio-economic stratum.

In a study in Nigeria, as many as 71 per cent of rural dwellers have reported inappropriate HSB during their last illness episode while only 53 per cent of urban dwellers reported inappropriate HSB during their last illness episode. Similarly, Nigerian women living in areas where the ratio of population to Primary Healthcare Centre (PHC) was high (more than 9,000:1) were less likely to have a skilled birth attendant present during childbirth than areas where the ratio of population to Primary Healthcare Centre (PHC) was lower (less than 6,000:1).

There are numerous variables, such as age, sex, level of education, culture, religion, past experience, that have a serious impact on our actions regarding health problems. In terms of age, people from different age groups take the disease differently, infants and young children have different behaviours than adults. Secondly, several publications in the field of gender and health have demonstrated that there are gender differences with regard to the decision-making process surrounding the proper form of care and treatment. For example, some studies have shown that women in developing countries use formal health care to a lesser degree than men and are more inclined toward traditional healing options instead. Studies investigating gender gaps in the perception and presentation of symptoms show that women not only ask for more social support than men, but also record substantially higher rates of distress. Education has been described as one of the most significant variables affecting health indexes. General awareness that individuals get from formal education or literacy such as in schools are more inclined to have a proper and appropriate behaviour towards health issues. As they get aware about particular diseases and take necessary steps for its improvement. Furthermore, the role of culture and religion have a major impact on the ways of treating an illness, specifically in Indian society where culture and religion strongly prevail. However, cultural and religious factors many a time gives rise to difficulty in dealing with the treatment of an illness. Individuals also treat the illness from their past experiences, they tend to use the same treatment which benefited them in the past and avoid the ineffective methods from the past. Nonetheless, marital status has its own effects on taking care of an illness, married women tend to face more issues with their health as compared to married men. This is one stigma that persists to be there in the society towards the diseased patients with certain illnesses that makes them feel aloof and detached due to the negative and unfavorable opinions of the community members and this leaves an impact on minds of the patients regarding their treatment. It includes not only negative opinions but the positive as well. Other than this the availability of medical services and treatment has another major role as the more easily available and accessible medical centers make individuals more encouraged to

visit and get proper treatment. Although, the unavailability of inaccessibility of these medical services demotivate and halt the patients to get the treatment, this, on the whole happens in rural areas. Not to mention, the trusted services and health providers promote and reassure the patients to take the treatment properly whenever it is needed instead of neglecting or avoiding.

But, even though some patients tend to avoid the symptoms and signs of the illness have to realize the necessity of visiting the medical center once the symptoms start to grow severe, people usually avoid the illness until the signs of it begin to show its severity and once these symptoms start affecting their ordinary life and it becomes difficult to meet the daily basic needs, that is when they start taking care of them and move for the better treatment. They continue to remain under the treatment of their illness as long as the symptoms persist to exist that not only makes them feel frightened but hopeless if it continues with the no or slow improvement. The longer the symptoms remain the more intolerant such patients become. On the other hand, the knowledge about the illness plays a vital role, if the patient is aware about the symptoms of the illness, he tends to visit sooner or later, often or rarely, gets serious or carefree regarding the treatment. However, if the illness is incurable or fatal, the patient loses hope and tends to show aggressive and detached behaviour. Although, in other cases, some of the patients start viewing the remaining time in an optimistic manner. There have always been differences in opinions and conduct regarding health and its treatment.

Check Your Progress: 2

- Note:
- 1) Use the space provided below for your answers.
 - 2) Compare your answers with those given at the end of the unit.
1. List the dominant variables that play role in health seeking behaviour.

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5.4 IMPACT OF RURAL HEALTH SERVICES

India's National Rural Health Mission (NRHM), one of the largest public health programs in the world launched in 2005, increased the uptake of institutional delivery and antenatal services, particularly among the poorest socio-economic groups, in the less-developed high-focus Indian states. Although the uptake of antenatal services were not improved in the early post-NRHM period 2007–08, there were considerable increase in the uptake of, and decline in its inequity in the late post-NRHM period 2011–12.

Larger equity impacts in the uptake of institutional delivery and antenatal services were found in the late post-NRHM period 2011–12 than in the early post-NRHM period 2007–08, indicating that public health programmes in developing country settings will have larger equity impacts after its almost full implementation and widest outreach. Impact of NRHM was greatest in those

states with higher proportion of beneficiaries enrolled under conditional cash-transfer programme of Janani Suraksha Yojana.

Studies found that NRHM's JSY payments were associated with increases in health facility births and decline in neonatal mortality, improvement in immunization rates and breastfeeding practices, decline in economic inequality in institutional delivery in the districts of higher JSY coverage, and decline in maternal mortality in richest districts than in the poorest.

ASHA services have connected marginalised communities with the health care system. Substantial variations in the receipt of ASHA services were reported with 66 per cent of women in northeastern states, 30 per cent in high-focus states, and 16 per cent of women in other states. In areas where active ASHA activity was reported, the poorest women, and women belonging to scheduled castes and other backward castes, had the highest odds of receiving ASHA services.

The role of the AYUSH system in providing health care services in rural India is palpable, with an abysmally inadequate health infrastructure. In rural India, the grossly inadequate health staff are greatly replenished by AYUSH doctors and paramedics. Many of the therapeutics are used in various ways that are safe and efficient for treating community health issues. Many of the concepts mentioned in the classical Ayurveda texts and other medical systems such as Yoga and Naturopathy are being used and many of them are proposed in the implementation and planning of the state programme (PIP).

The aim of rural health services is on improving the health of individuals living in rural India. They are attempting to make health facilities available to this disadvantaged population. It has already had no impressive impact on the improvement of health care services.

5.5 HEALTH SEEKING BEHAVIOUR DUE TO TECHNOLOGY

The number of internet users has increased considerably worldwide. The internet is used for health purposes by the general public, and the importance of the internet as a source of health information is growing. Additionally, health-related websites have gained popularity increasingly. Internet-based health information is accessed from a variety of sources, including websites run by organizations, homepages run by individuals, and online support groups where people actively exchange health information and blogs. Online health information has been giving opportunity to consumers and health care professionals for interactive health communication approach. Health care seekers appear to be using the internet to enhance their medical care; they report using the information online to diagnose a problem and feel more comfortable about their health provider's advice given the information found on the internet.

Access to reliable disease information online has been linked to reduced anxiety, increased feelings of self-efficacy, and decreases in utilization of ambulatory care. Therefore health-related websites have the potential to

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powerfully influence the attitudes and behaviour of consumers. The advantage of the internet is that it is widely available (home, work, libraries), convenient (24 hours a day at home) and anonymous. Health information has permeated the internet. Electronic health information is becoming ubiquitous. Health information is one of the most frequently sought topics on the Internet. The amount of health information available online is astounding. Using the most common method consumers rely on to find healthcare information online a keyword search on “health” at a popular search engine recently generated 473,000,000 results.

In the last decade, the percentage of adults who have accessed the Internet to look for health information has rapidly increased. In a national survey in the Netherlands in 2012, 81 per cent of people aged 65-75 years used the Internet and 54 per cent used the Internet for health information. Research shows that online health consumers tend to be more educated, earn more, and have high-speed internet access at home and at work.

Reuters reported that on average 53 per cent of Americans search the internet for health information and, according to the Pew Internet and American Life Project (2003), of the 63 per cent of Americans who access the Internet (128 million people), 66 per cent of these look for health and medical information. Eysenbach estimated that, on a global level, of the 278 million internet searches that are being conducted each day, approximately 12.5 million searches are health related. Internet users searching for health information will even go to sites aimed at health professionals. Understanding who is more likely to use the internet as a health information tool is an important aspect of understanding how the internet is transforming health care.

However, some have drawn attention to the dangers of patients using the internet for health information. For example, some raise the potential for misdiagnosis and exploitation. Others suggest that internet use can erode patients’ faith in the authority of health-care practitioners. In response to such concerns, health-care providers have established classificatory systems for evaluating the scientific worth of web information. It is important that health professionals acknowledge patients’ search for knowledge, that they discuss the information offered by patients and guide them to reliable and accurate health websites. Therefore, health-care professionals need to improve their own skills in Internet use.

Check Your Progress: 3

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. Elaborate on the role of technology in shaping rural health seeking behaviour.

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5.6 ALTERNATIVE MEDICINE AND RURAL HEALTH

Alternative Medicine as a term used to define a group of diverse medical and health care systems, practices, and products that are not generally considered part of conventional medicine (medicine as practiced by holders of MD or DO degrees and by their allied health professionals such as physical therapists, psychologists, and registered nurses). Well-known examples of CAM include aromatherapy, naturopathy, osteopathy, chiropractic, yoga, massage, acupuncture, and herbal and nutritional supplements. While other terms, including “integrative,” “alternative,” “unconventional,” and “holistic” are often used to describe this group of treatments.

Available data on geographical differences in CAM consumption indicate that the prevalence of CAM use in rural regions is higher than in urban areas. A longitudinal study on women’s health in Australia found that women who consult alternative health practitioners were more likely to reside in nonurban areas, and an interview Study of 160 urban and 105 rural residents in Portugal revealed 75 per cent of the rural respondents as using CAM treatments and therapists while only 25 per cent of the urban respondents did so. Lind et al conducted an analysis of 2,37,500 insurance claims from 2 large insurance companies in the United States and found that the proportion of claimants using chiropractors was higher amongst ruralbased residents when compared to those claimants in urban areas (although users of chiropractic in urban areas had more chiropractic visits than users of chiropractic in rural areas). As these findings suggest, there may well be differences between the types of CAM used across geographical location.

AYUSH, an acronym for Ayurveda, Yoga and Naturopathy, Unani, Siddha, Sowa- Rigpa and Homeopathy represents the alternative systems of medicine recognized by the Government of India. Traditional medicine is defined as an amalgamation of knowledge, skill, and practices based on theories, beliefs, and experiences indigenous to different cultures, whether explicable or not, used for therapeutic, restorative, prevention, diagnosis and maintenance of physical and mental health. These systems are recognised globally for complementing disease prevention, treatment and generic health maintenance. There is widespread use of traditional medicine across developing countries (across Asia, Africa and Latin America) with rapidly emerging markets in North America and Europe. However, despite increasing national and international attention, the formal health systems, particularly in resource-poor settings, are yet to harness its true potential. Recognising such intricacies, the 67th World Health Assembly resolution on traditional medicine has been instrumental in the development of updated WHO Traditional Medicine Strategy (2014–23) with objectives to harness its contribution and promote effective use.

These international developments are of particular significance for India that has a pluralistic medical culture with a well-documented history and practice of alternative medicinal forms namely—Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homoeopathy—now jointly referred to as AYUSH. As integral part of Indian culture, Indian System of Medicine (ISM) comprising of Ayurveda, Yoga,

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Unani and Siddha was practiced even before formal health system took shape. Unfortunately, in spite of presenting an effective role for health maintenance (preventive and curative), the ISM has been neglected and marginalized since the pre-independence era (before 1947). Prejudiced approach like curbing the state patronage, abolition of ISM schools and creation of medical bureaucracy further undermined the legitimacy of ISM practice. The potential of ISM was further curbed by the perils of caste-, class-, communal-and language-based politics. Besides, strong epistemological preferences within ISM system undermined its scope and expansion as a formal health care system in the post- independence era. However, time and again, with the realization of strengthening comprehensive primary health care, various Expert Groups and Health Committees in India have recommended integration of ISM with the biomedical/ allopathic system. For instance, Mudaliar Committee was first to recommend utilizing indigenous doctors for delivering vertical healthcare programs. Subsequent national health policies such as National Health Policy (1983), National Education Policy in Health Sciences in 1989 and National Health Policy (2002) have also pointed out the potential of ISM in improving healthcare access, particularly in the absence of modern healthcare in rural India. But attempts made towards its revival thereafter, only saw bio-medicalization of these systems either for validation through scientific methods or for commercialization and excessive marketization of indigenous therapies. However, the paradigm shift in government's policy post-independence has focused on: creation of schools for ISM to produce licentiates, standardizing curriculum, drug research for patenting and mass production. The institutionalization of ISM in keeping with international interests further led to the creation of the Department of Indian Systems of Medicine and Homeopathy (ISM&H) in 1995 (later renamed as Department of AYUSH in 2003). In pursuance of 'mainstreaming' policy, post-2005, National Rural Health Mission (NRHM) provided an opportunity where strategies like provisioning of AYUSH drugs, co-locating providers at public health facilities and inter-sectoral convergence with ISM functionaries implementing national health programs were devised. Department of AYUSH also launched National AYUSH Mission during 12thlpan with an objective of providing affordable, sustainable and accessible care. Subsequently, the elevation of AYUSH Department into an independent Ministry (in November 2014) is a noteworthy policy decision to further upgrade AYUSH educational standards with emphasis on epistemological strengths, quality standardization and stewardship.

Check Your Progress: 4

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) What is alternative medicine?

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2) What is the role of AYUSH in improving health care in rural India?

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5.7 BARRIERS TO HEALTH SEEKING BEHAVIOUR

Individual Factors: Lack of personal resources, such as time and social support, and work-related factors such as job pressures, long working hours, and unemployment as barriers to HSB. Physical impediments such as illness and disability and negative personal characteristics such as low motivation, laziness, and lack of self-discipline can also adversely affect the HSB of an individual. A lack of financial resources is also a barrier to HSB.

Cultural factors: Perceived lack of consideration for cultural beliefs, traditions, and practices by health care professionals can cause stress and be a barrier to health service access.

Environmental factors: Aspects of the physical environment can act as a barrier for HSB. Differences in climate and seasonal temperatures can be a barrier for HSB. Sociocultural influences acting as a barrier to HSB included the availability, affordability, and easy access to fast foods.

Economic factor: The financial issue has always been one or another reason as a barrier for HSB. Particularly, in rural areas, the socio-economic conditions hinder towards the positive change in HSB.

Health Service Access: Lack of familiarity with the health care system in new place can act as a barrier to HSB, and the lack of information available in their original language. These barriers can act as impediments to accessing services, particularly for the migrants and the elderly.

Distance and lack of transportation: In many rural areas the medical centers are at a great distance which makes it difficult for the patient to get the treatment at the proper time. Not only the distance, but the lack of transportation adds up to the issue of getting the needed healthcare.

Lack of personnel as a barrier to health seeking behaviour was mainly reported by patients from the rural areas.

5.8 CHALLENGES IN RURAL HEALTH

The six decades of Indian independence have witnessed too many plans, papers and proposals giving top priority to the health issues in India. Unfortunately, amid huge economic growth, health remains the greatest predicament. Even WHO slogans 'Health for All,' 'Millennium Development Goals,' and more recently, 'Free Health Care' have not been converted into concrete action on the ground. Accessibility of healthcare as well as the use of available healthcare facilities, particularly in rural areas, continues to be low in India.

The poor state of health system in rural areas is a result of consolidated outgrowth of degraded system. It explains not only the distance between the existing policy and infrastructure but obstruction in development too. The expenditure on public health system has not only been ignored by the state but also by the common mass. People mostly prefer private practitioners and private hospitals over government run hospitals. The disillusionment and frustration with the growing ineffectiveness of the government sector is gradually driving poor people to seek help of the private sector, thus forcing them to spend huge sums of money on credit, or they are left to the mercy of 'quacks'. Therefore, it is very essential for us to review primary elements for degradation of Public health system in India.

Some of the challenges in rural health are:

- Inadequate human resources
- Inefficacious infrastructure
- Inclination towards Home Based Deliveries
- Lack of coordination between Medical Research and Health Service delivery Institutions
- High Infant Mortality
- Non-preparedness to fight with Epidemic in rural areas
- Apathetic attitude of medical professionals
- Dominance of unregulated Private medical professionals.

Poor quality services at state-run hospitals force many people to visit private facilities. The overall health care utilization is also low, only half of (52%) of Indian mothers receive three or more antenatal checkups; Only 43.5 per cent of children in India receive all vaccinations {(NHFS-3), 2005-06}. At one side our peripheral health centres are underutilized whereas on other side our tertiary and secondary (District) level facilities are often argued as overloaded with the work that could have been done at lower centres, resulting in compromise of quality. Others continue to argue that even these centres have not successfully exploited the skills of its specialists with only 2–3 OPDs/OTs per week per physician. The underutilization of peripheral centres is attributed to varied factors related to accessibility, quality, affordability, deficient human resource, poor monitoring, lack of community participation and ownership. Vast and diverse geographical locations of India inhibit proper penetration of health care delivery in such areas. Further, health care personnel are reluctant to work at block or below level areas, as they have to face two challenges, first the absence of reasonable living conditions (e.g. proper housing, 24 hour electricity supply, good school for their children, social isolation etc.) and second, the under functioning of majority of health care facilities in such areas and hence no opportunity to translate their technical skills. Absence of stringent transfer policy leads to frustration among staff. Posting of surgeons at under functioning facilities at the beginning of their career erodes their surgical skills and make them non-functional forever. The absence of accessible quality primary care services leads many poor people to either forgo medical care altogether or choose to seek expensive and unregulated care in the private sector. Lack of

adequate quality data on burden of disease and trauma for proper planning along with poor public health awareness are a few more issues. MCI and Nursing council of India in their current shape are inert to some extent as their main focus is only on quantitative assessment of staff, infrastructure, material and equipment rather than quality or treatment audit. The qualitative assessment e.g. professional skills of staff, managerial skills of administrators, quality of health care provided, quality of students trained at these institutes are totally missing. Deep rooted corruption is prohibiting the smooth flow of system, especially the prompt purchase of medical equipment and diagnostics. Directorate of Medical and Health, the apex administrative and regulatory body for Medical and Health in India lacks the technical expertise and needs overhauling. The middle level managers of health system e.g. CMHOs, BMOs are unable to accomplish tasks at their own level. They act as a weak link between higher authority and periphery. System has taken away their self-esteem.

Check Your Progress: 5

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) What are the barriers and challenges to health seeking behaviour?

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5.9 SOLUTIONS TO RURAL HEALTH

The RURBAN initiative of developing villages can be gainfully used for innovative medical manpower management in primary healthcare. Thus, the concept of Model Group Housing at block level/PHC level should be considered where government employees of all the departments e.g. Medical and Health, PWD, School Education, Police, Electricity, Bank, JaldayaVibhag, BDO, Road and Transport, Post and Telegraph etc. could be provided accommodation where required. Facilities like school, playground, community centre, supermarket etc. could be provided in the neighbourhood. This would take care of the “Doctors Deficiency” argument very often put forward as an excuse for non-availability of medical and health facilities. This concept would allow holding, retaining and recruiting fresh talent by facilitating their stay and improvising their quality of life comparable with their counterparts in the city. Thus, the feeling of being deprived and frustrated could be compensated.

The critics might argue that it is a very optimistic project and shall require long time to complete while consuming a lot of resources. While one has waited for solutions including conditional provisions of rural posting, increased allowances or even making the rural services compulsory to doctors or other skilled workers, for more than several decades, the project looks worth serious consideration. The resources under RURBAN model, the funds from Prime Minister exchequer, Member of Parliament, Village Development Project and those under the National Rural Health Mission with low cost housing along with several other projects could be merged to give impetus to the newer solutions.

The model housing township should also harbor the first referral unit (FRU) consisting of a gynecologist, anesthesiologist, pediatrician and surgeon with facility of ICU. Such team effort would provide cohesive and coordinated medical services. The primary health centre physicians could also stay at the model township and may be allowed to run the OPDs and the National health programmes or other specific responsibilities by a 'to and fro' movement every day. The initial phase of populating such hospitals in such model villages (or townships) can happen by way of temporary deployment of skilled manpower from larger govt hospitals or tertiary care centres that feed such rural areas. Emergencies at the PHCs can be either handled through ambulance 108, or the mobile surgical services stationed at the model township which can periodically conduct camps and handle necessary medical, surgical, emergency and blindness control programme. The sub centres could remain connected to FRU by the ASHA worker or the incharge, who could be from the AYUSH staff. The proposal to effectively mainstream AYUSH in present rural health delivery systems would go a long way and also pave way for looking into the deep rooted indigenous system of medicine well accepted among people. There could be value addition of services by AYUSH staff if they are trained for national health programmes, identifying emergencies in facilitating a decrease in IMR, MMR and IFR. Needless to state, the huge work force of AYUSH is available with majority of the states in India. Further, to reduce the out of pocket expenses, innovative insurance model combining the RSBY with "TOP UP" Scheme created by UNICEF and is proposed by the author as means to bring change under his leadership.

The instant health advice 'electronically' proposed by NIDAN would add to the quality healthcare. Similarly, either outsourcing the diagnostics based on the Rajasthan CT/MR PPP Model could not only add to the quality results and management but would also provide public awareness in the rural areas for early diagnosis and planned preventive strategies. The PPP diagnostic model along with the innovative insurance scheme can generate huge financial resources (the State Government being partner) for the development both at primary and secondary care services. Further, the model townships could be connected through telemedicine to the tertiary care centres for availability of speciality/ superspeciality consultation and also continuing medical education. The health managers could be made administrative incharge for procurement and maintenance of equipment and on a day to day basis could also organize financial planning systems of accountability thereby relieving the medical professionals for delivering their professional services at ease. Paramedic training and orientation could be a periodic phenomena with the Health managers sustaining efficient health delivery system.

Alternatively, measures like pooling of locally available private specialist doctors or sharing specialist from one PHC to another or if there is a lack of manpower at the FRU outsource the overall health services either to the local nursing homes or private cooperative of doctors or the corporate hospitals could be mobilized. The NABH or JCI implementation with regional modifications could improve the quality of services.

To give the impetus to the whole new concept the administrative machinery needs to be integrated and reorganized. The Medical, Health and Education

Department need to work in synergy to achieve the objective of overall enhancement of health. It is, therefore, possible for three Departments to be supervised by a singular Principal Secretary. The convergence link for primary, secondary and tertiary level health services should be Director General of Health Services. There could be five Directors under DG namely Director, Infectious Diseases and Epidemiology (including statistics); Director, Non communicable and Life Style Based Diseases (including Nutrition); Director, Mother and Child Health and Family Welfare; Director, Training, continuous Medical Education, School Health and IEC; Director, Mobile Health Unit, Transport, Communication and Information Technology. These could cohesively interact with other supportive departments like PHED and Science and Technology. To give impetus to research on endemic and perennial diseases, the establishment of “Model Rural.

Township’ concept of ICMR could be very useful. The State Medical Councils (presently inert) could assert their authority in pushing and updating the professional’s knowledge by using the human resources at tertiary and secondary care. The tertiary care should be given the task of conducting epidemiological surveys to identify regional rural health issues. All these efforts could create a healthy milieu for rural health practices.

Finally, provision of healthcare for rural areas hinge on the affordability of treatment and diagnostic costs. In order to propel the indigenous production of medical devices, drugs, surgicals and diagnostics, the biomedical scientists in the hospitals, research institutions and elsewhere can come together and translate their knowledge into affordable medical products. By instituting ‘innovation clinics’, the consulting scientists and doctors could join hands in order to translate their respective knowledge useful for bedside of patient around the Model Rural Research Centre of ICMR. This will be important in fulfilling the concept of ‘make in India’ thereby saving both the costs of import thereby enabling affordable health care.

5.10 LET US SUM UP

Considering the fact that over 65 per cent of the country’s population still lives in rural areas where the health infrastructure, awareness and education levels are very poor, planning for health care is all the more difficult. Since independence, numerous programmes have been initiated, including recent introduction of AYUSH and traditional medicine. However, it is imperative to understand the health seeking behaviour as it dominantly determines the success or failure of policies and programmes at the ground level.

5.11 KEY WORDS

Malnutrition refers to deficiencies, excesses, or imbalances in a person’s intake of energy and/or nutrients.

Alternative Medicine as a term used to define a group of diverse medical and health care systems, practices, and products that are not generally considered part of conventional medicine (medicine as practiced by holders of MD or DO degrees and by their allied health professionals such as physical therapists, psychologists, and registered nurses).

AYUSH, an acronym for Ayurveda, Yoga and Naturopathy, Unani, Siddha, Sowa- Rigpa and Homeopathy represents the alternative systems of medicine recognized by the Government of India.

Traditional Medicine is defined as an amalgamation of knowledge, skill, and practices based on theories, beliefs, and experiences indigenous to different cultures, whether explicable or not, used for therapeutic, restorative, prevention, diagnosis and maintenance of physical and mental health^{43,44}.

Healthcare Seeking Behaviour (HSB) has been defined as, “any action or inaction undertaken by individuals who perceive themselves to have a health problem or to be ill for the purpose of finding an appropriate remedy”. Health seeking behaviour can also be referred to as illness behaviour or sick-term behaviour.

5.12 FURTHER READINGS

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15.13 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Refer to Section 5.2

Check Your Progress: 2

1. Refer to Section 5.3

Check Your Progress: 3

1. Refer to Section 5.5

Check Your Progress: 4

1. Refer to Section 5.6
2. Refer to Section 5.6 pg 76

Check Your Progress: 5

1. Refer to Section 5.8

